

COMPUTER NETWORKS SYLLABUS

UNIT I

Introduction: Uses of Computer Networks, Types of networks: WAN, LAN, MAN, Network Topologies, Reference models: OSI, TCP/IP.

Physical Layer: Transmission media: magnetic media, twisted pair, coaxial cable, fiber optics, wireless transmission.

UNIT II

Data link layer: Design issues in data link layer: framing, flow control, error control, Error Detection and Correction: Parity, CRC checksum, Hamming code, Flow Control: Sliding Window Protocols, Applications : Data link layer protocols HDLC, PPP.

UNIT III

Medium Access sub layer: Channel allocation problem, MAC Protocols: ALOHA, CSMA, CSMA/CD, MAC addresses, IEEE 802.X, Standard Ethernet, Wireless LANS. Bridges, Types of Bridges.

UNIT IV

Network Layer: Design issues in Network Layer, Virtual circuit and Datagram subnets-Routing algorithm: Shortest path routing, Flooding, distance vector routing, Link state routing, Hierarchical routing, Broad casting, Multi casting, Routing for mobile hosts.

Internetworking: Concatenated Virtual Circuits, Connectionless internetworking, Tunneling, Internetwork routing, Fragmentation.

UNIT V

Network layer in internet: IPv4, IP addresses, Sub netting, Super netting, NAT.

Internet control protocols: ICMP, ARP, RARP, DHCP, Network Layer in ATM Networks.

Congestion Control: Principles of Congestion, Congestion Prevention Policies.

Congestion Control in datagram Subnet: Choke packet, load shedding, jitter control.

Quality of Service: Leaky Bucket algorithm and token bucket algorithm.

UNIT VI

Transport Layer: Transport Services, Connection establishment, Connection release and TCP and UDP protocols.

Application Layer: Domain name system, FTP, HTTP, SMTP, WWW

TEXT BOOKS

1. Computer Networks — Andrew S Tanenbaum, 4th Edition. Pearson Education/PHI
2. Data Communications and Networking – Behrouz A. Forouzan. Third Edition TMH.
3. Data Communication and Networks- , Bhushan Trivedi, Oxford Higher Education.

REFERENCES

1. An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition, Pearson Education
2. Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson

Chapter 1

Introduction

What is Computer Network

- A collection of Autonomous computers interconnected by any media.
- Two computers are said to be **interconnected** if they are able to exchange information.
- The connection may be via a copper wire/ fiber optics/microwaves/infrared/ satellites.

Why people are interested in Computer Networks

- Resource sharing:

To make all physical resources such as printers, scanners, equipment, etc., available to anyone on the network without regard to the physical location of the resource and the user.

- Information sharing:

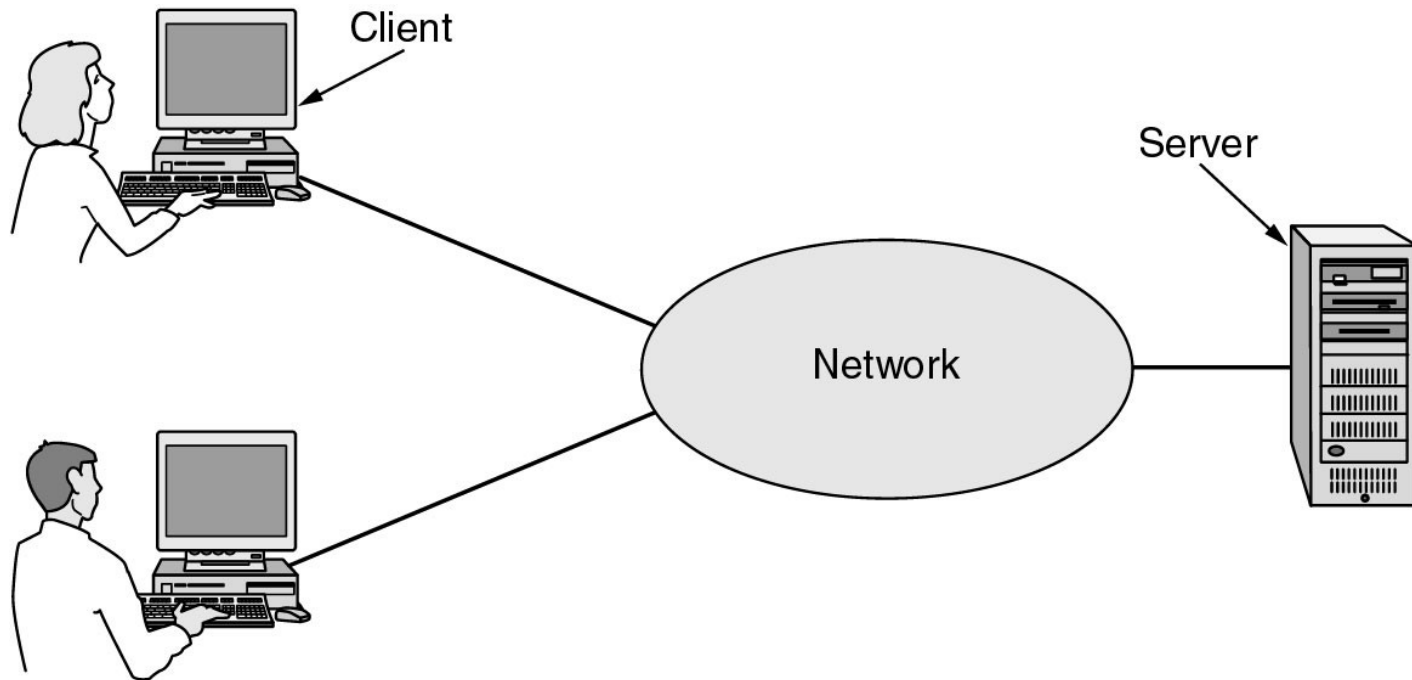
Allowing users to access relevant information and documents, i.e data available to any one on the network instantly without regard to the location of the data and the user accessing it.

Uses of Computer Networks

- Business Applications
- Home Applications
- Mobile Users
- Social Issues

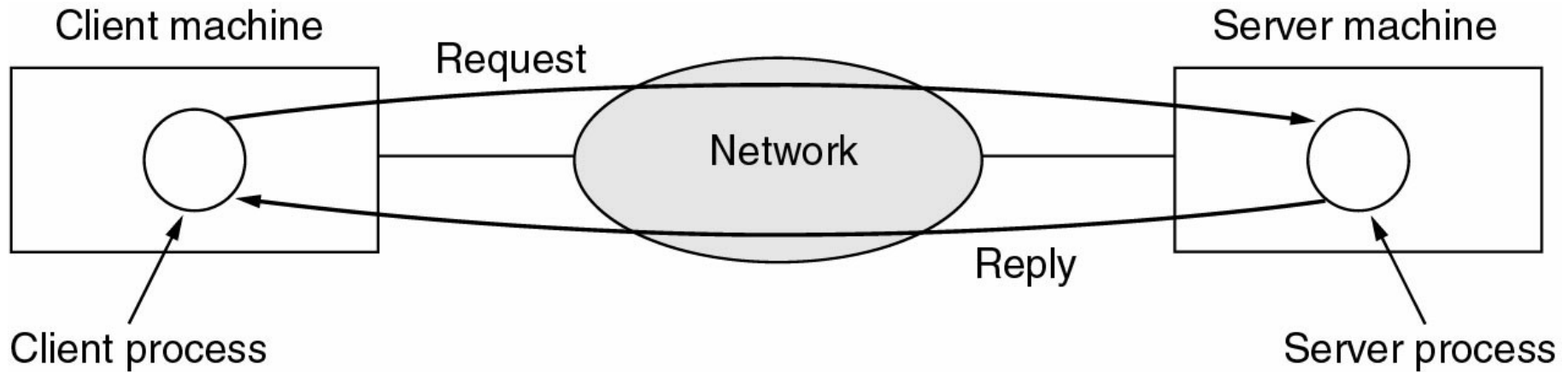
Business Applications of Networks

- A network with two clients and one server.



Business Applications of Networks (2)

- The client-server model involves requests and replies.

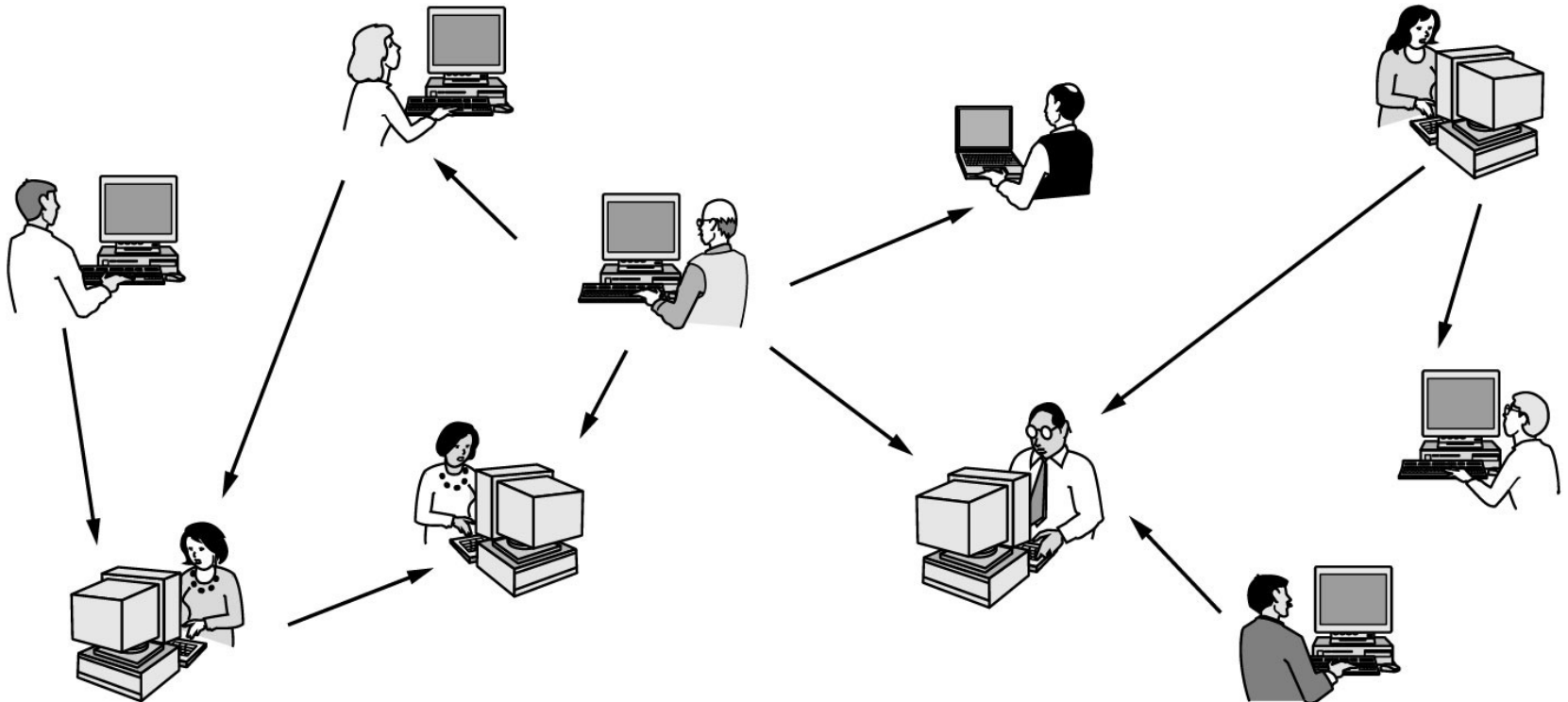


Home Applications

- Access to remote information
- Person-to-person communication
- Interactive entertainment
- Electronic commerce

Home Applications (2)

- In peer-to-peer system there are no fixed clients and servers.



Home Applications (3)

- Some forms of e-commerce.

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing

Mobile Users

- Combinations of wireless networks and mobile computing.

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Types of transmission technology

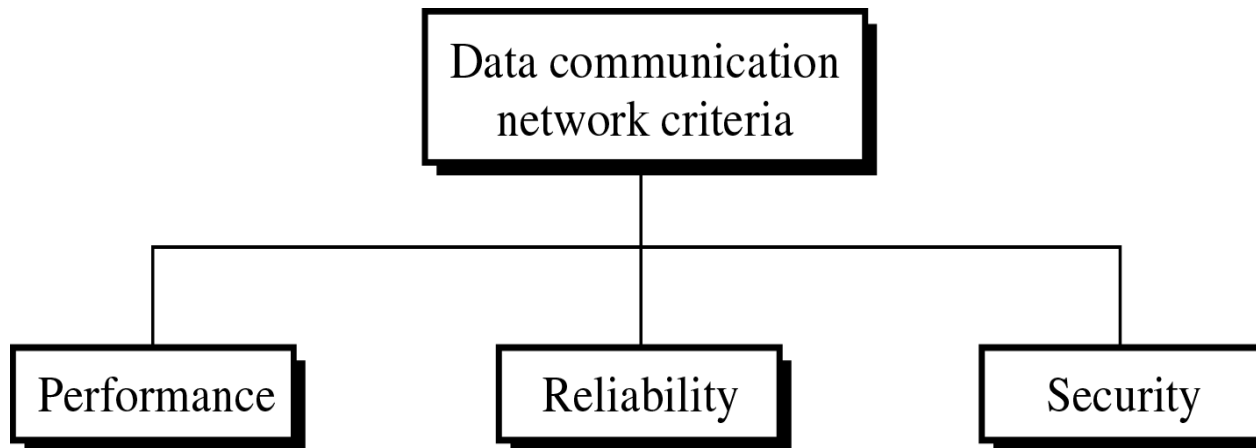
- Broadcast links
- Point-to-point links

Contd..

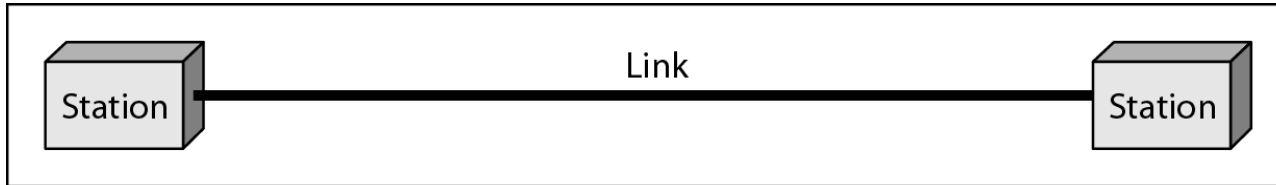
- Broadcast networks have a single communication channel that is shared by all the machines on the network.
- Short messages, called packets, sent by any machine are received by all the others. An address field within the packet specifies the intended recipient.
- Broadcast systems allow possibility of addressing a packet to all destinations by using a special code in the address field.
- Some broadcast systems support transmission to a subset of the machines, known as multicasting.
- To indicate that it is multicasting one bit is reserved and remaining $n-1$ address bits can hold a group number.
- In contrast, Point-to-Point networks consists of many connections between individual pairs of machines.
- Point-to-Point transmission with one sender and one receiver is called Unicasting,

Network

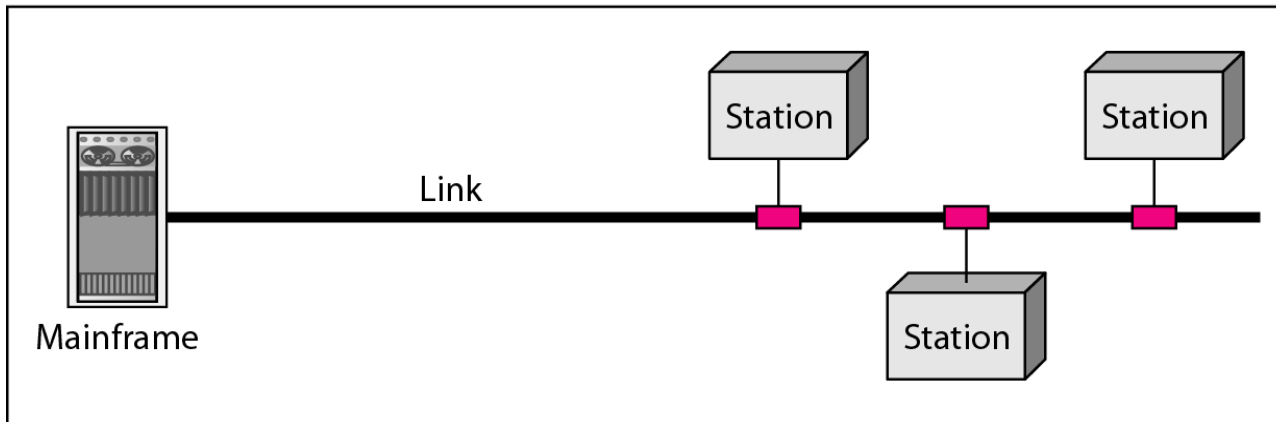
- Network: A set of devices (nodes) connected by communication links
- Node: Computer, printer, or any device capable of sending and/or receiving data
- To be considered effective and efficient, a network must meet a number of criteria



Type of Connection



a. Point-to-point

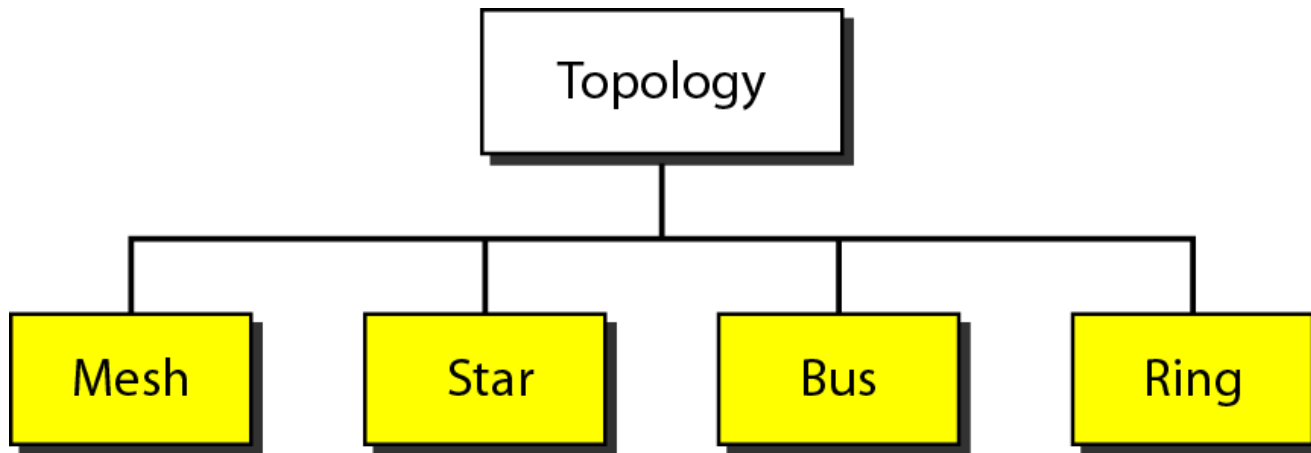


b. Multipoint

Type of Connection

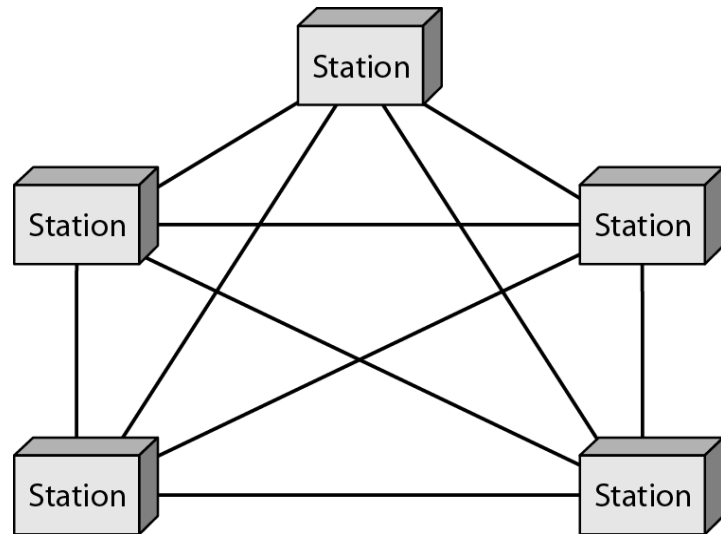
- Point-to-point
 - Dedicated link between two devices
 - The entire capacity of the channel is reserved
 - Ex) Microwave link, TV remote control
- Multipoint
 - More than two devices share a single link
 - Capacity of the channel is either
 - *Spatially shared*: Devices can use the link simultaneously
 - *Timeshare*: Users take turns

Physical Topology



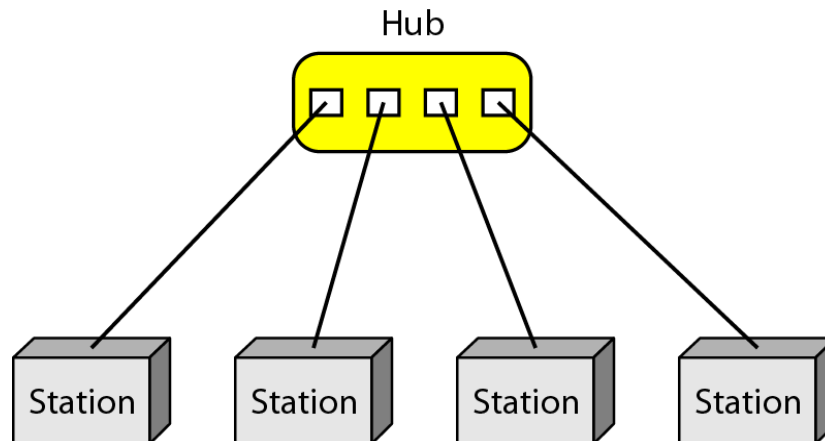
Mesh Topology

- Dedicated point-to-point link to every other nodes
- A mesh network with n nodes has $n(n-1)/2$ links. A node has $n-1$ I/O ports (links)
- Advantages: No traffic problems, robust, security, easy fault identification & isolation
- Disadvantages: Difficult installation/reconfiguration, space, cost



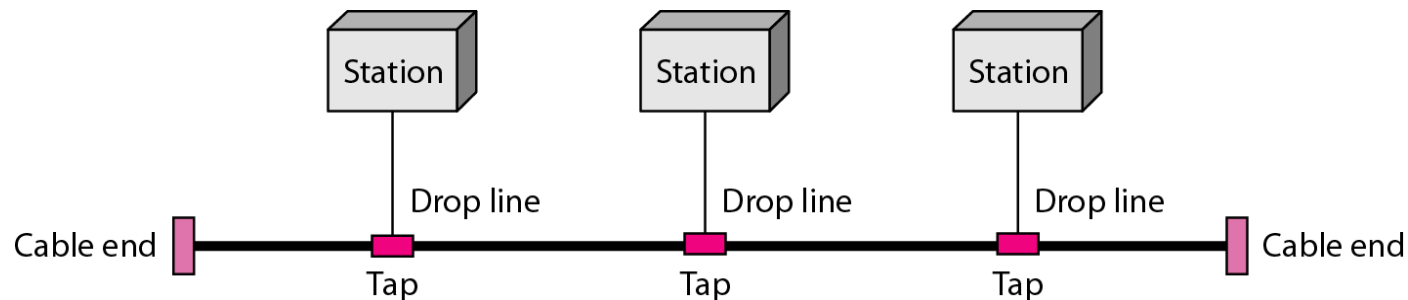
Star Topology

- Dedicated point-to-point link *only to* a central controller, called a **hub**
- Hub acts as an exchange: No direct traffic between devices
- Advantages: Less expensive, robust
- Disadvantages: dependency of the whole on one single point, the hub



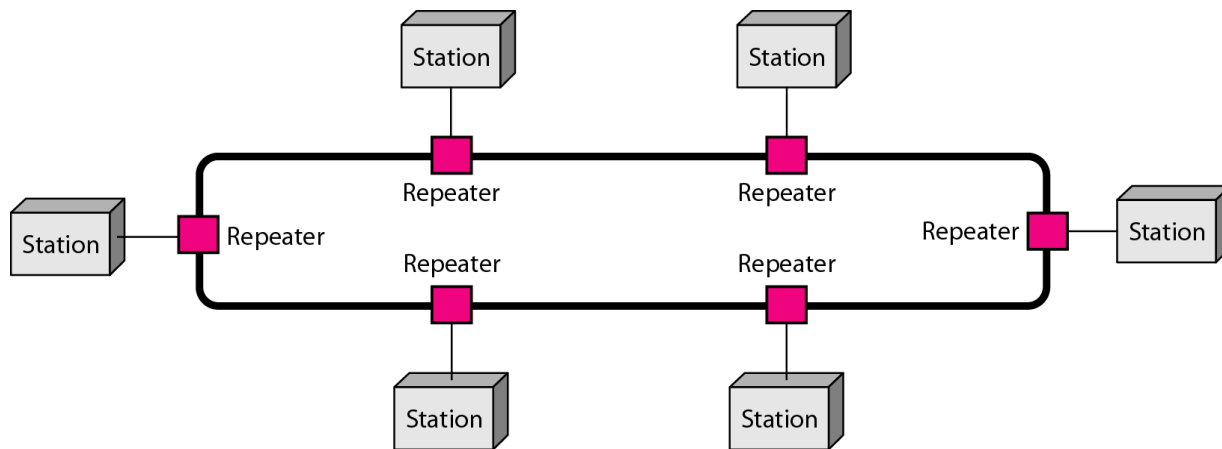
Bus Topology

- One long cable that links all nodes
- tap, drop line, cable end
- limit on the # of devices, distance between nodes
- Advantages: Easy installation, cheap
- Disadvantages: Difficult reconfiguration, no fault isolation, a fault or break in the bus stops all transmission



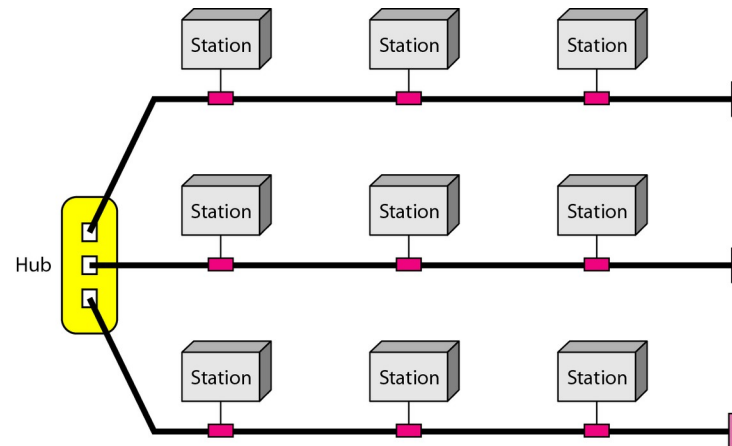
Ring Topology

- Dedicated point-to-point link only with the two nodes on each sides
- One direction, repeater
- Advantages: Easy reconfiguration, fault isolation
- Disadvantage: Unidirectional traffic, a break in the ring can disable the entire network



Hybrid Topology

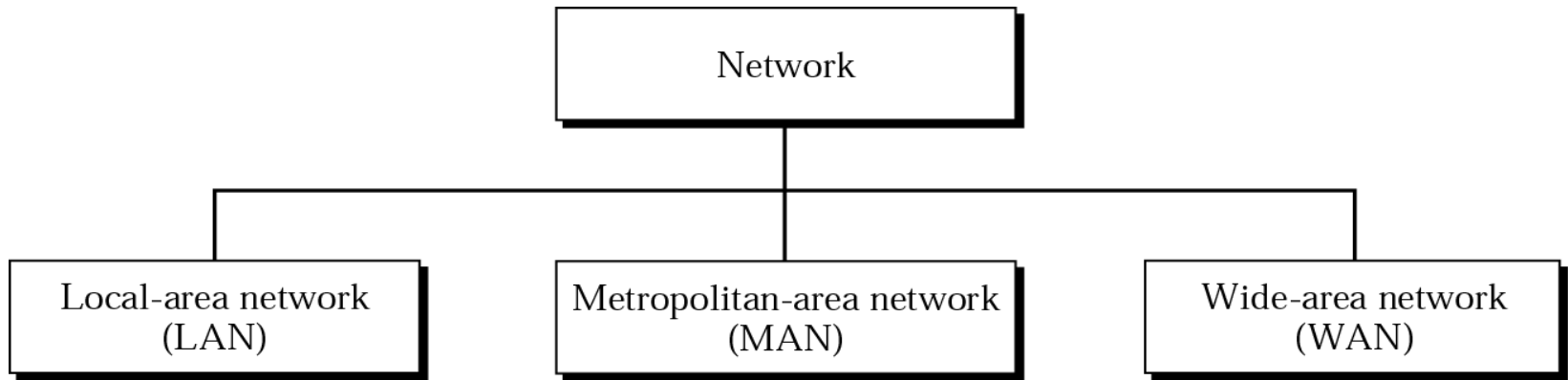
- Example: Main star topology with each branch connecting several stations in a bus topology
- To share the advantages from various topologies



Types of Network

- Local Area Networks
- Metropolitan Area Networks
- Wide Area Networks

Categories of Networks

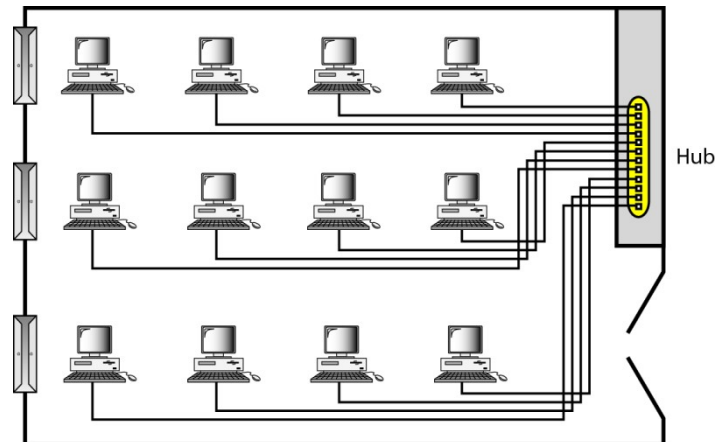


Classification of interconnected processors by scale

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

LAN (*Local Area Networks*)

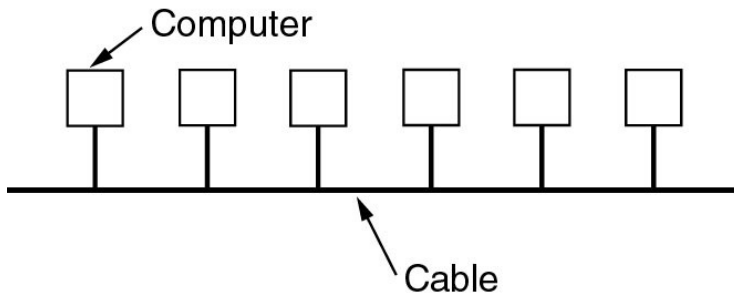
- Usually privately owned
- A network for a single office, building, or campus $\leq$ a few Km
- Common LAN topologies: bus, ring, star
- An isolated LAN connecting 12 computers to a hub in a closet



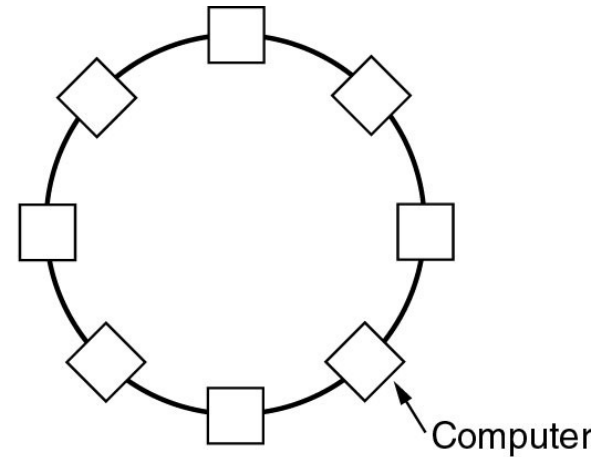
LAN contd..

- LANs, are **privately-owned networks** within a **single building or campus of up to a few miles in size**.
- To connect personal computers and workstations in company offices and factories to share resources (e.g. printers) and exchange information.
- LANs are distinguished from other kinds of networks by three characteristics: **1) their size, 2) their transmission technology, and 3) their topology**.
- Restricted in size
- Use a transmission technology consisting of a single cable to which all the machines are attached, like the telephone company party lines once used in rural areas.
- **Traditional LANs run at speeds of 10 to 100 Mbps**, have low delay (tens of microseconds), and make very few errors.

Local Area Networks



(a)



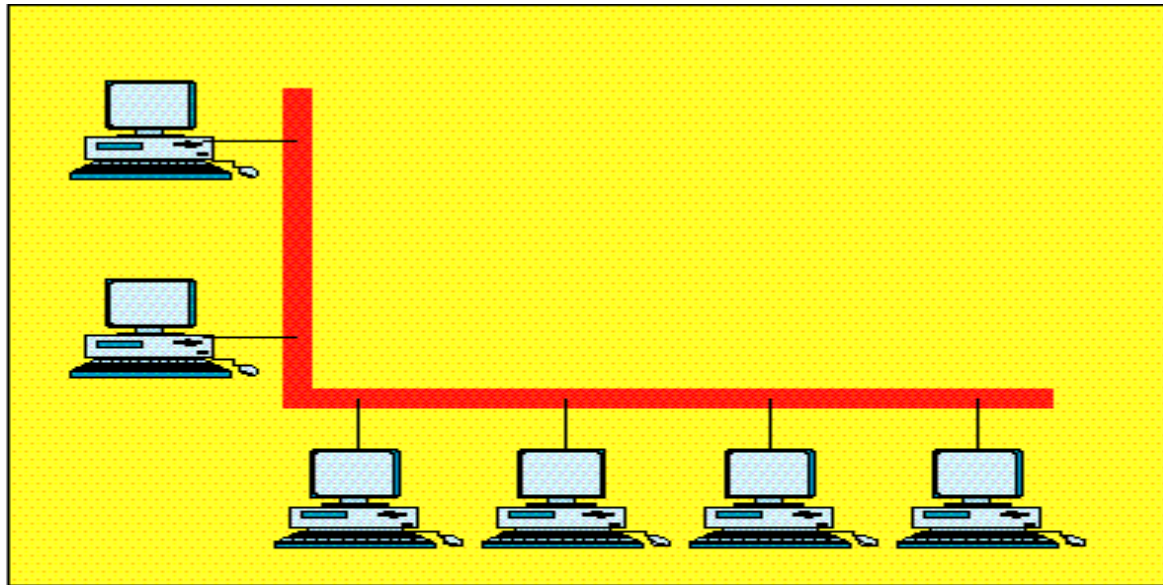
(b)

- *Two broadcast networks*
- *(a) Bus*
- *(b) Ring*

Contd..

- In a bus network, at any instant at most one machine is the master and is allowed to transmit. All other machines are required to refrain from sending.
- An arbitration mechanism is needed to resolve conflicts when two or more machines want to transmit simultaneously. The arbitration mechanism can be centralized or distributed.
- IEEE 802.3, called Ethernet, is a bus-based broadcast network with decentralized control, usually operates at 10 Mbps to 10 Gbps.
- The second type of broadcast system is the ring. In a ring, each bit propagates around on its own, not waiting for the rest of the packet to which it belongs.
- IEEE 802.5(the IBM token ring), is a ring-based LAN operating at 4 and 16 Mbps. FDDI is another example of a ring network.

LAN (Local Area Networks)



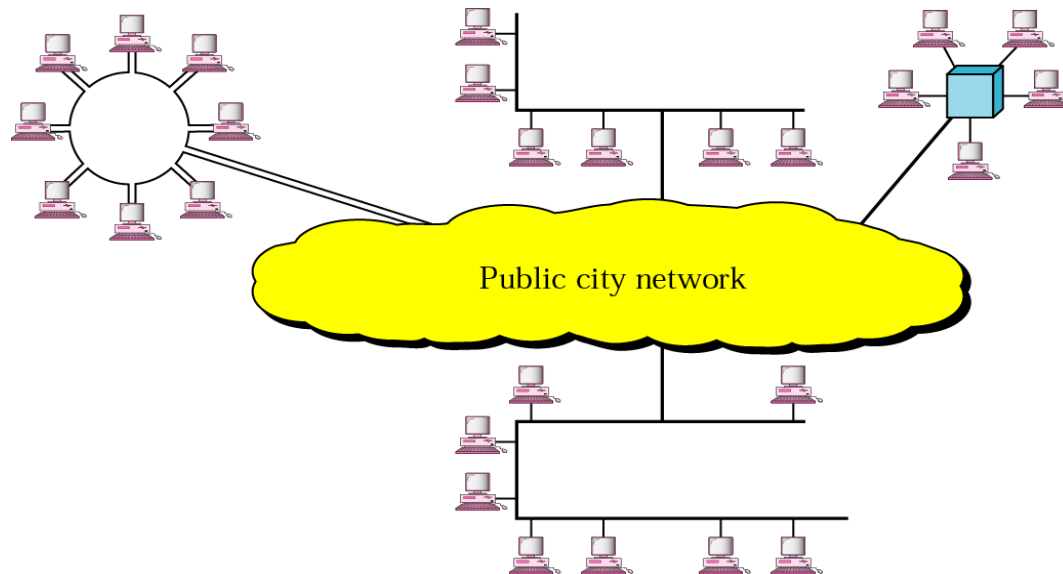
Single building LAN

Contd..

- Broadcast systems can be further divided into static and dynamic , depending on how the channel is allocated.
- A static allocation would be to divide time into discrete intervals and use a round-robin algorithm , allowing each machine to broadcast only when its time slot comes up. Static allocation wastes channel capacity when a machine has nothing to say during its allocated slot.
- Dynamic allocation can be either centralized or decentralized.
 - In the centralized channel allocation, there is a single entity, for e.g. a bus arbitration unit, which determines who goes next.
 - In decentralized channel allocation method, there is no central entity; each machine must decide for itself whether to transmit.

MAN (*Metropolitan Area Networks*)

- Designed to extend to an entire city
- Cable TV network, a company's connected LANs
- Owned by a private or a public company

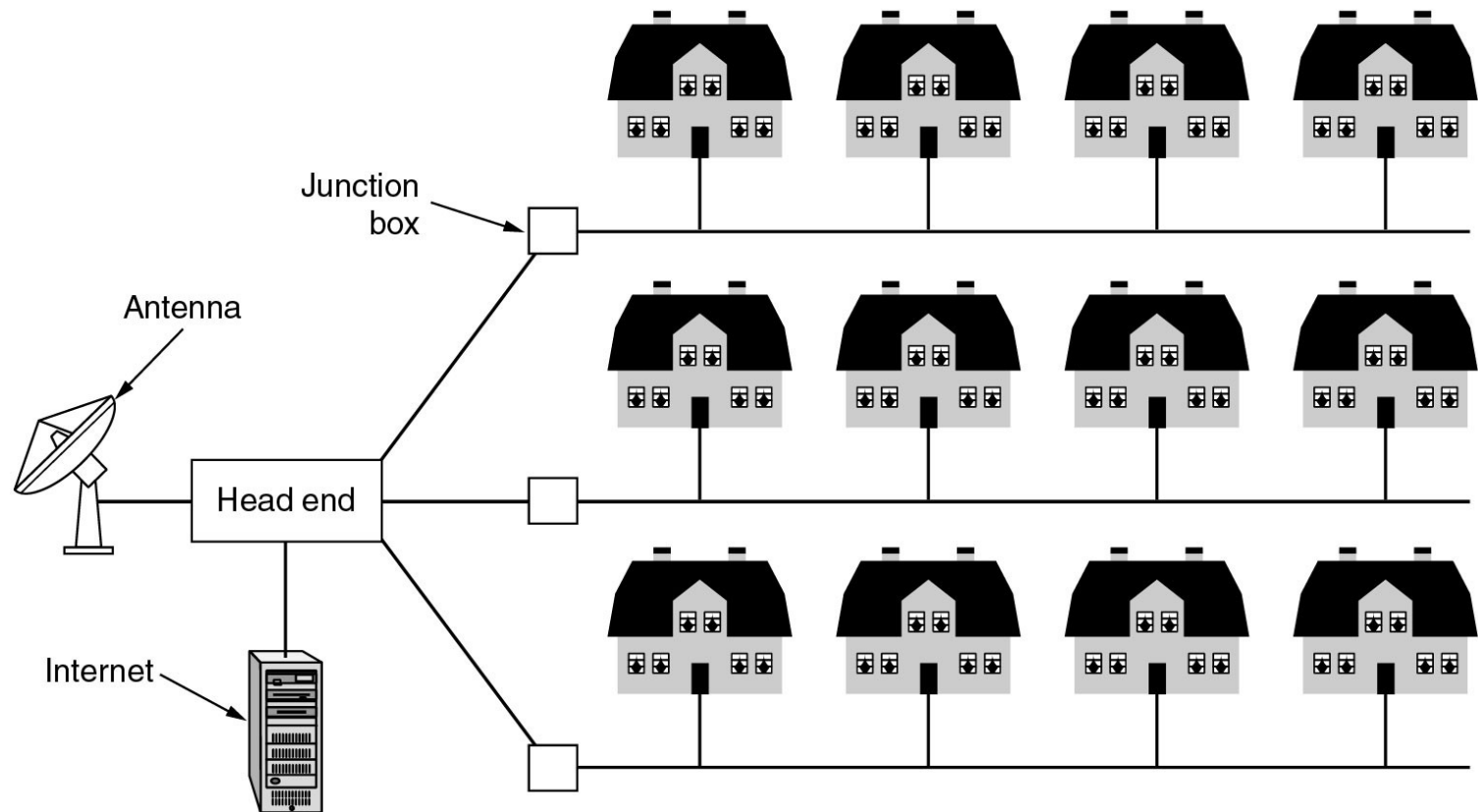


MAN (contd...)

- It is a bigger version of LAN
- Ex: Local Cable Television Network.
- It supports 802.6(IEEE standard) called DQDB (Distributed Queue Dual Bus)
- DQDB consists of two unidirectional buses(cables) to which all computers are connected.
- A metropolitan area network (MAN) is a network that interconnects users with computer resources in a geographic area or region larger than that covered by even a large local area network ([LAN](#)) but smaller than the area covered by a wide area network ([WAN](#)).
- city-10km

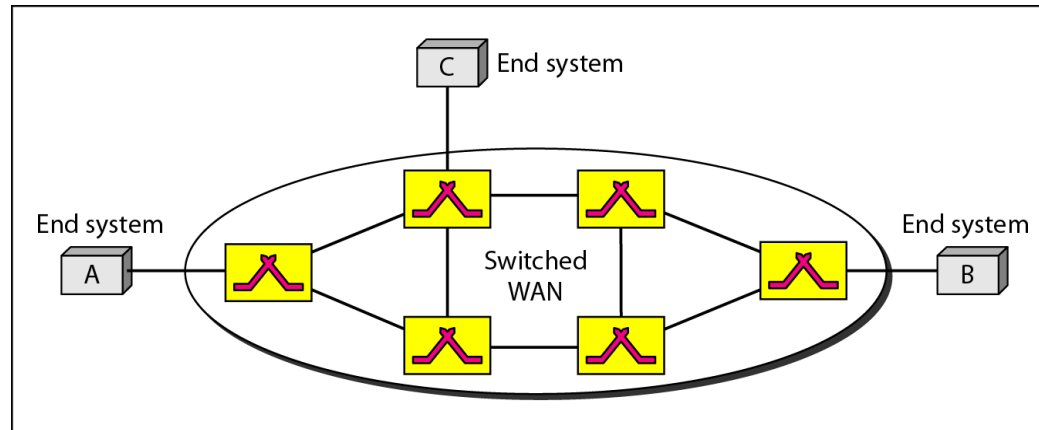
Metropolitan Area Networks

- A metropolitan area network based on cable TV.

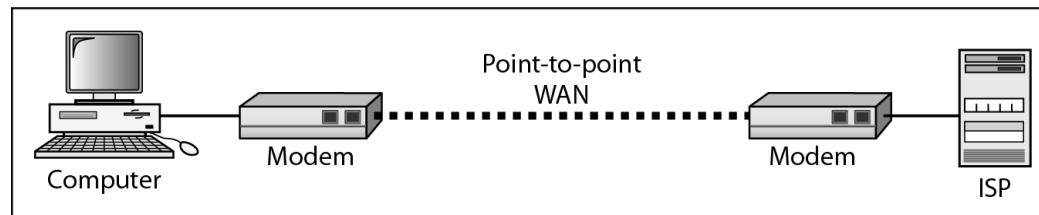


Wide Area Network(WAN)

- Long distance transmission, e.g., a country, a continent, the world
- Enterprise network: A WAN that is owned and used by one company



a. Switched WAN



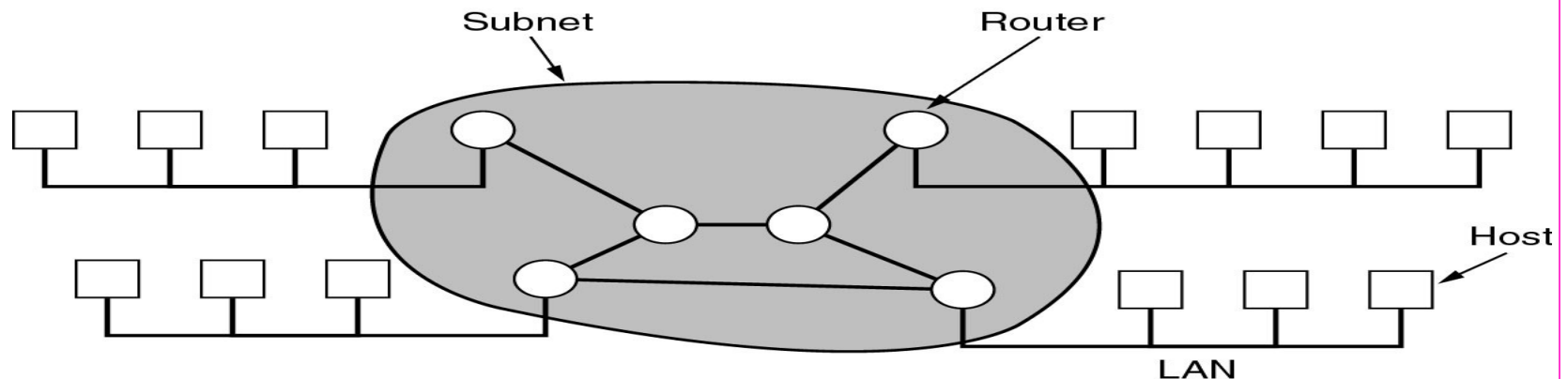
b. Point-to-point WAN

WAN contd...

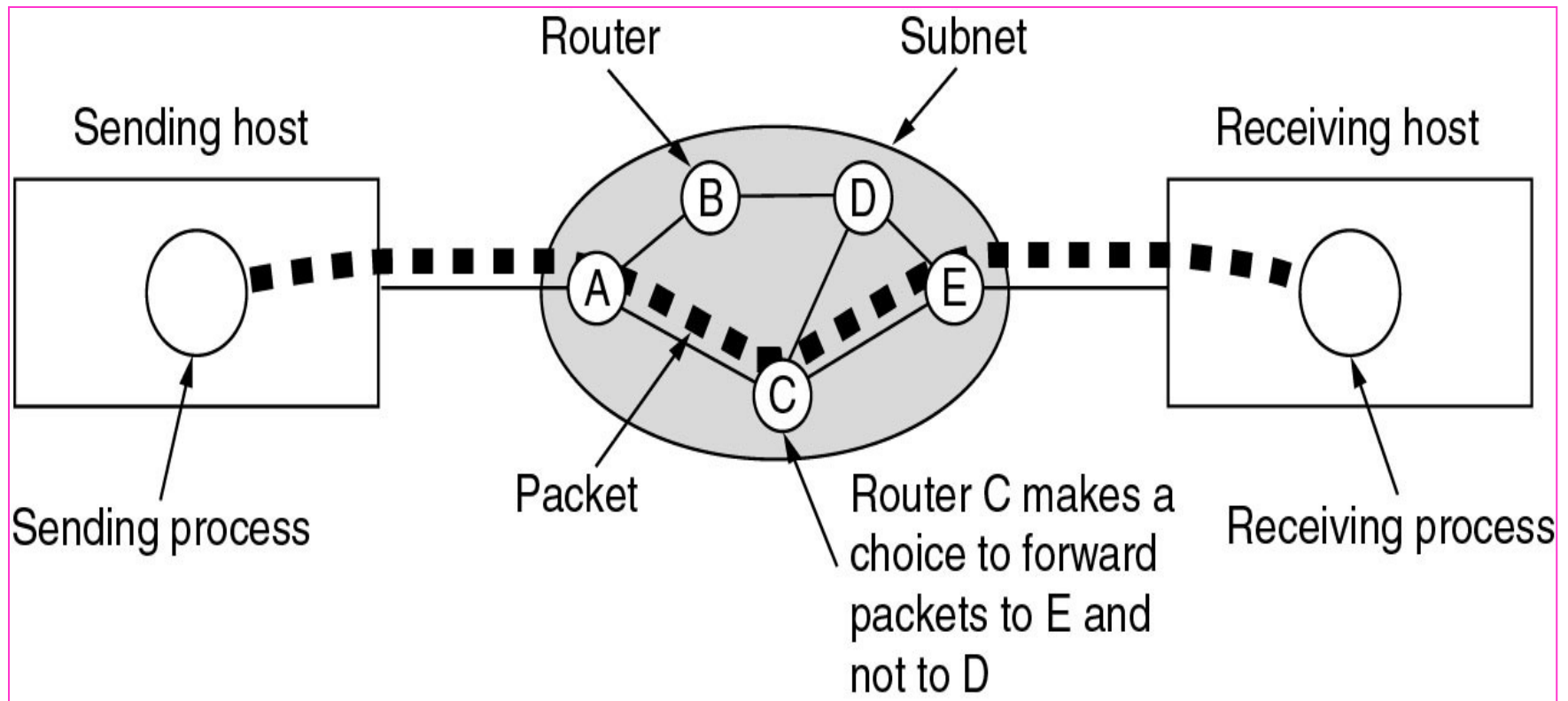
- *Network spread geographically (country or across globe) is called WAN.*
- *WAN contain hosts these are connected by a communication subnet.*
- *The job of the subnet is to carry messages from host to host.*
- *Ex: The telephone system carries words from speaker to listener.*
- *Relation between hosts and the subnet*
- *Country : 100 km and 1,000 km*

Subnet contains two distinct components

- *Transmission lines: (trunks, circuits, or channels)*
 - *move bits between machines.*
- *Switching elements:*
 - *Specialized computers used to connect two or more transmission lines.*

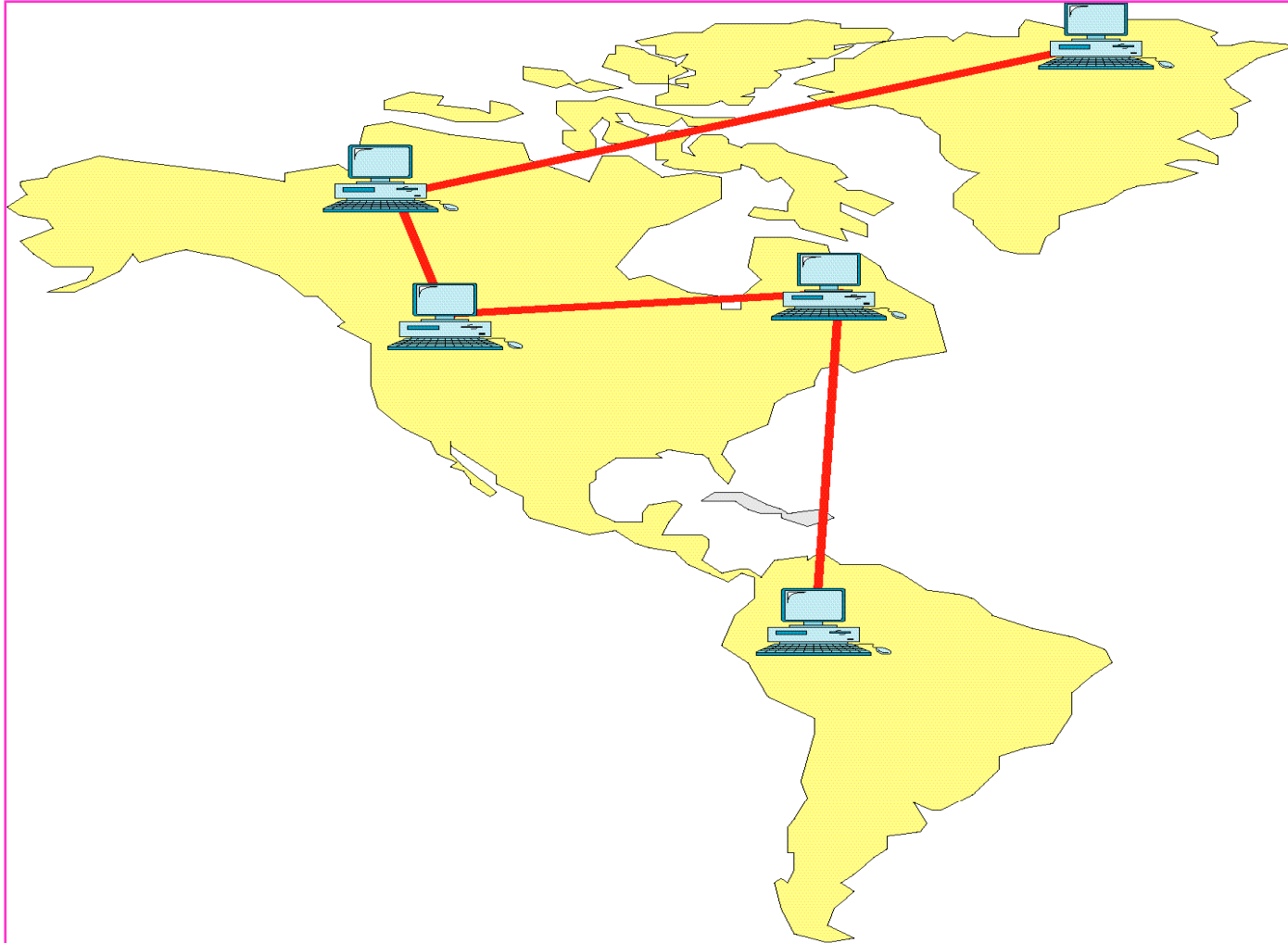


A stream of packets from sender to receiver



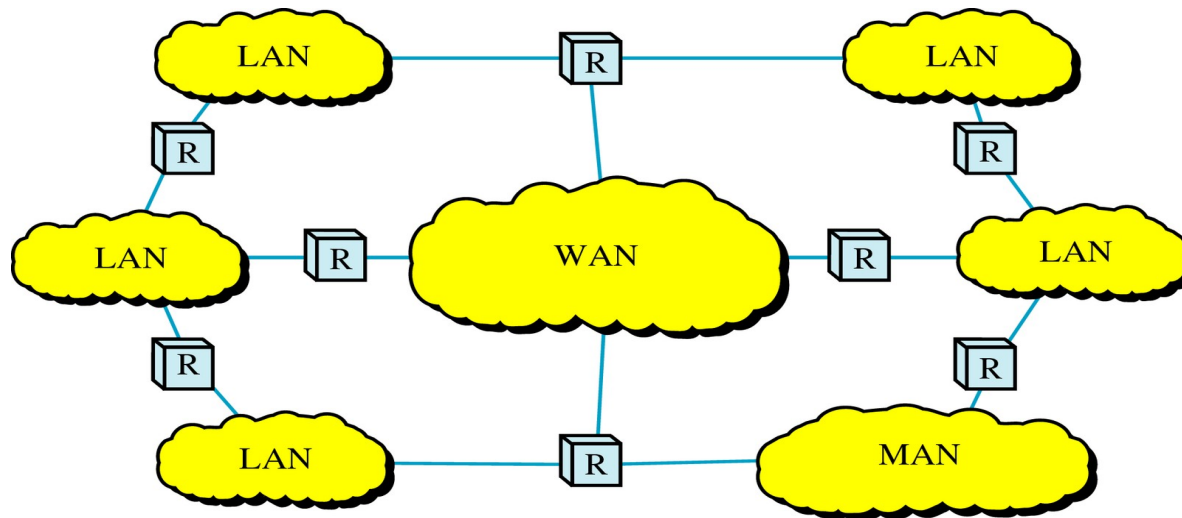
Point to point, store-and-forward, or packet-switched

Wide Area Network



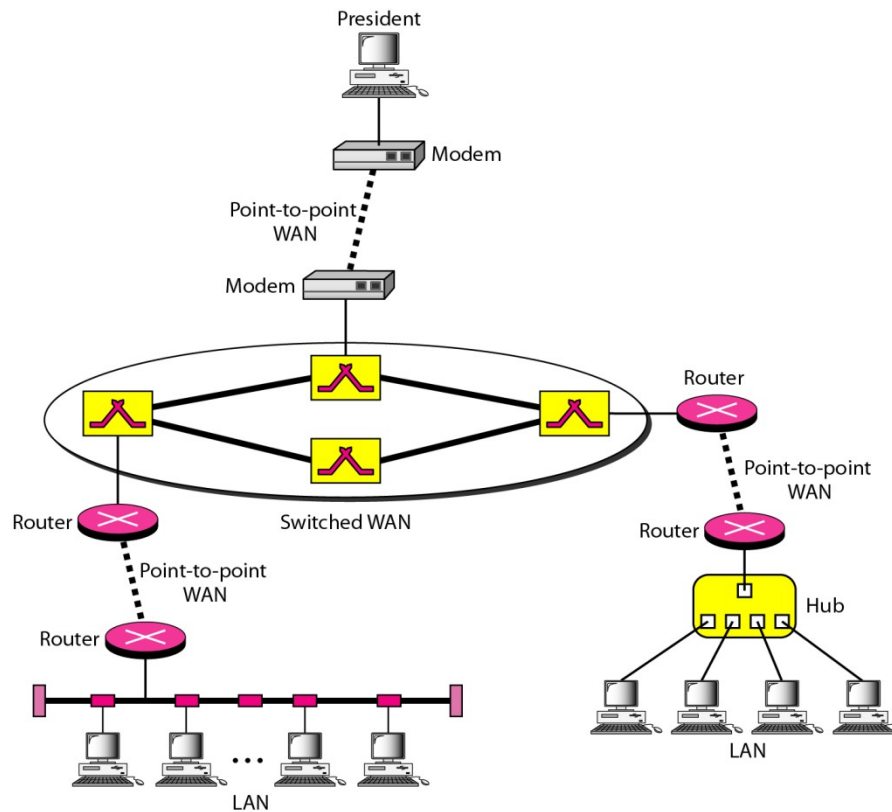
Internetwork

- Internetwork (internet) : two or more networks are connected by internetworking devices
- Internetworking devices: router, gateway, etc.
- The Internet: a specific worldwide network



Internetwork Example

- A heterogeneous network : four WANs and two LANs



Network Software

- Protocol Hierarchies
- Design Issues for the Layers
- Connection-Oriented and Connectionless Services
- Service Primitives
- The Relationship of Services to Protocols

Protocol Hierarchies

- To reduce the design complexity, most networks are organized as a stack of layers or levels, each one built upon the one below it.
- The number of layers, name of the layers, function of the layers differ from one network to other.
- Each Layer is to offer certain services to the higher layers.
- Each layer shields the details of how the offered services to the higher layers

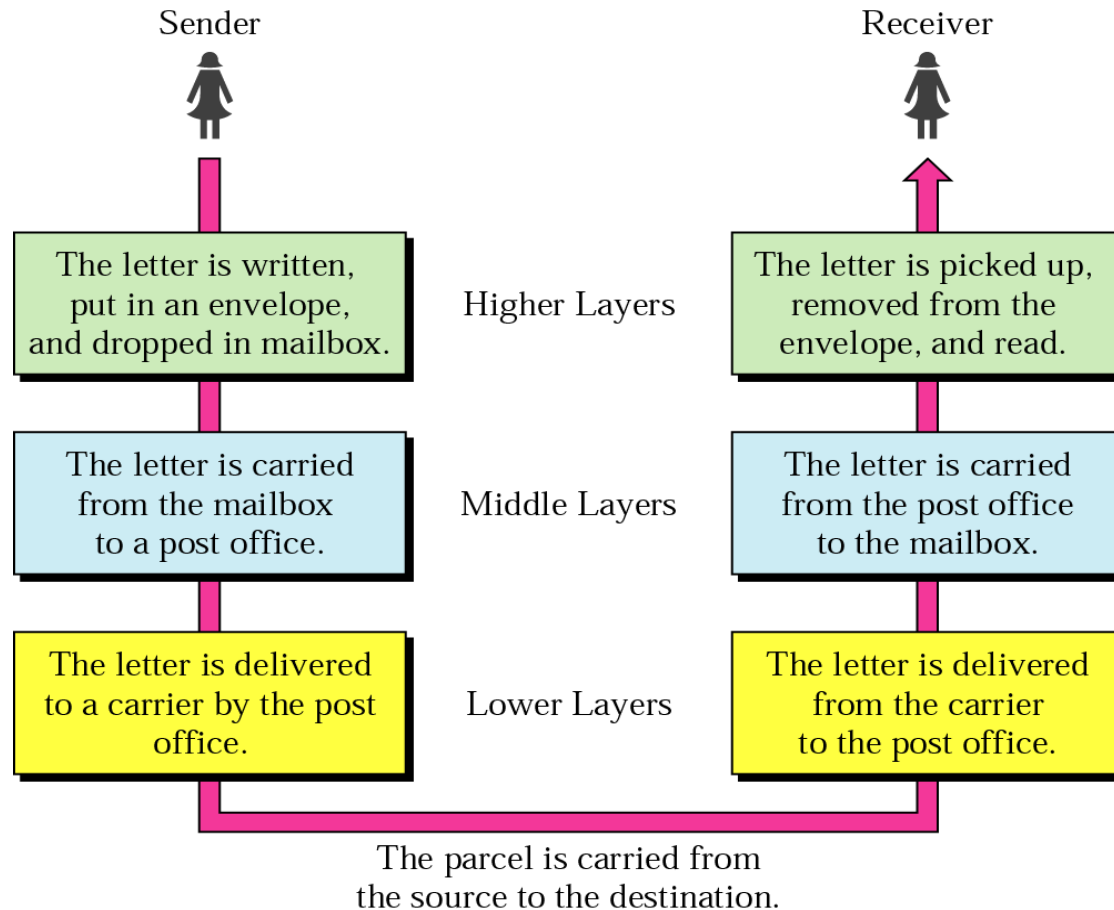
Technical words

- Protocol : The rules and conventions used in the conversation between layer ***n*** on one machine to layer ***n*** on another machine are collectively known as the layer ***n*** protocol.

A **Protocol** is an agreement between the communicating parties on how communication is to proceed.

- Peers: The entities comprising the corresponding layers on different machines are called **peers**.
- Interface: Between each pair of adjacent layers is an interface. The **interface** defines which primitive operations and services the lower layer makes available to the upper one.
- Network Architecture: A set of layers and protocols is called a network architecture.
- Protocol stack: A list of protocols used by a certain system, one protocol per layer, is called a **protocol stack**.

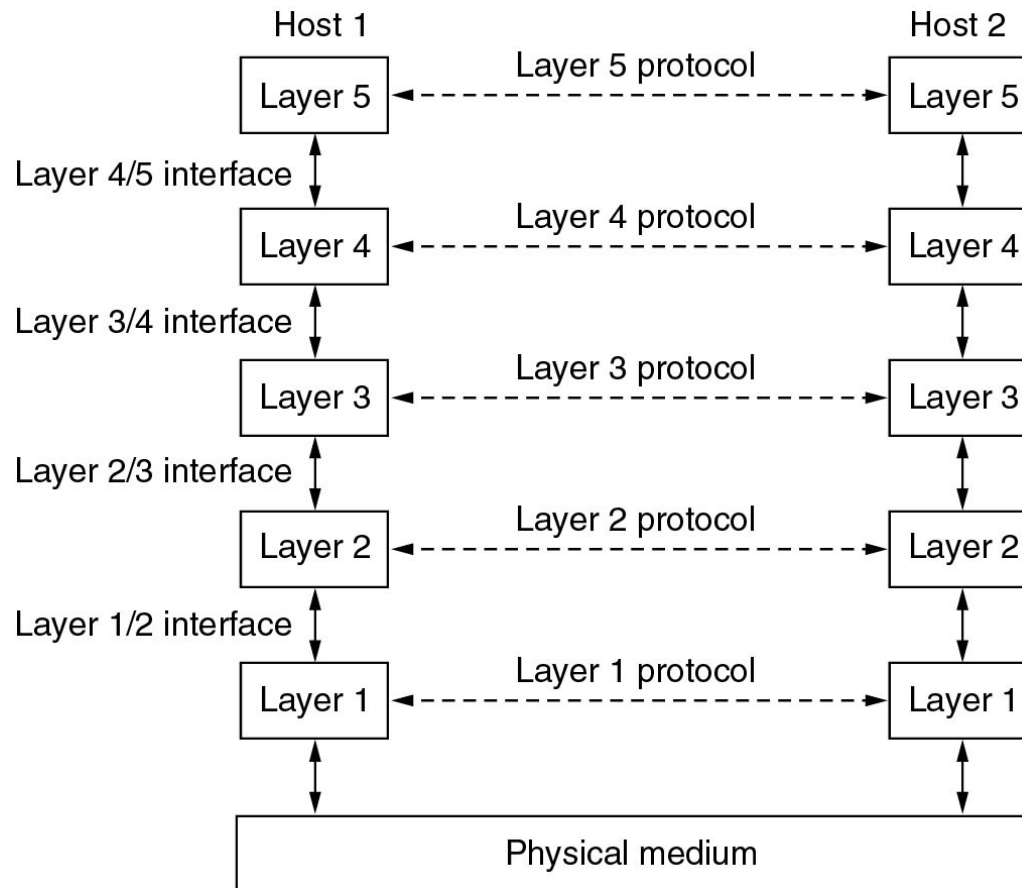
Layered Model: Sending a Letter



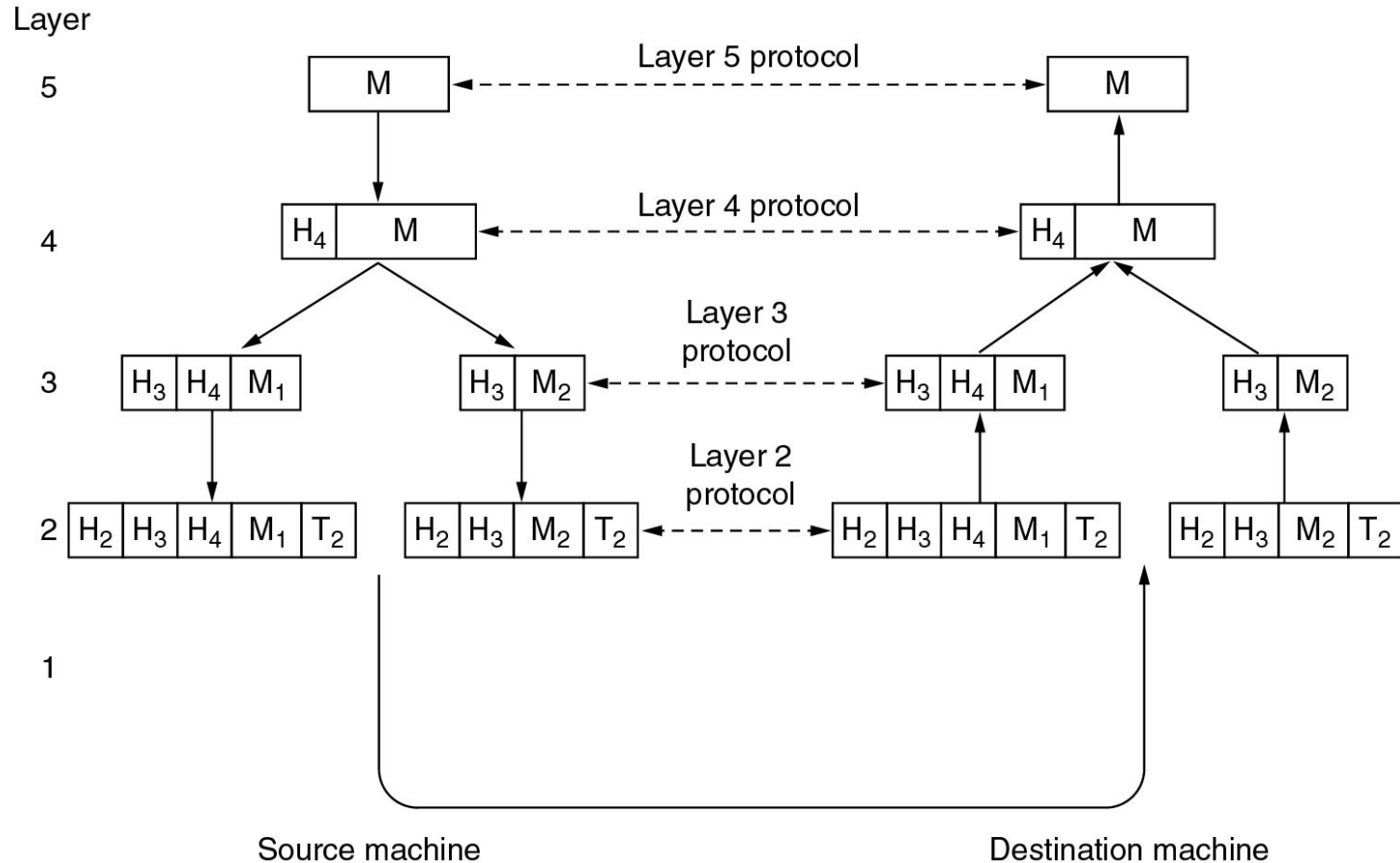
Network Software

Protocol Hierarchies

Layers, protocols, and interfaces.



Protocol Hierarchies (3)



Example information flow supporting virtual communication in layer 5.

Design Issues for the Layers

- Addressing
- Rules for data transfer
- Error Control
- Flow Control
- Disassembly and Reassembly
- Multiplexing
- Routing

Connection-Oriented and Connectionless Services

Six different types of service.

		Service	Example
Connection-oriented	{	Reliable message stream	Sequence of pages
		Reliable byte stream	Remote login
		Unreliable connection	Digitized voice
Connection-less	{	Unreliable datagram	Electronic junk mail
		Acknowledged datagram	Registered mail
		Request-reply	Database query

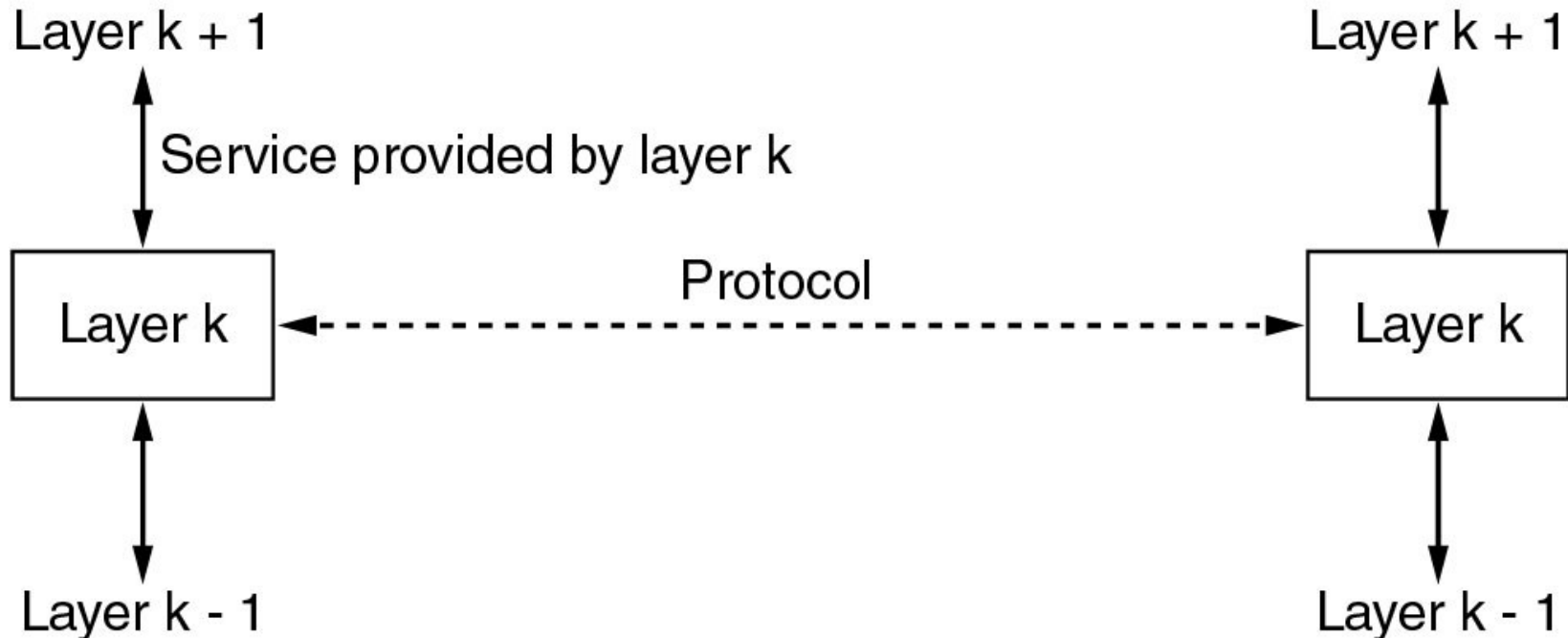
Service Primitives

- A service is specified by a set of primitives(operations) available to a user process to access the service.
- Five service primitives for implementing a simple connection-oriented service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Services to Protocols Relationship

- The relationship between a service and a protocol.

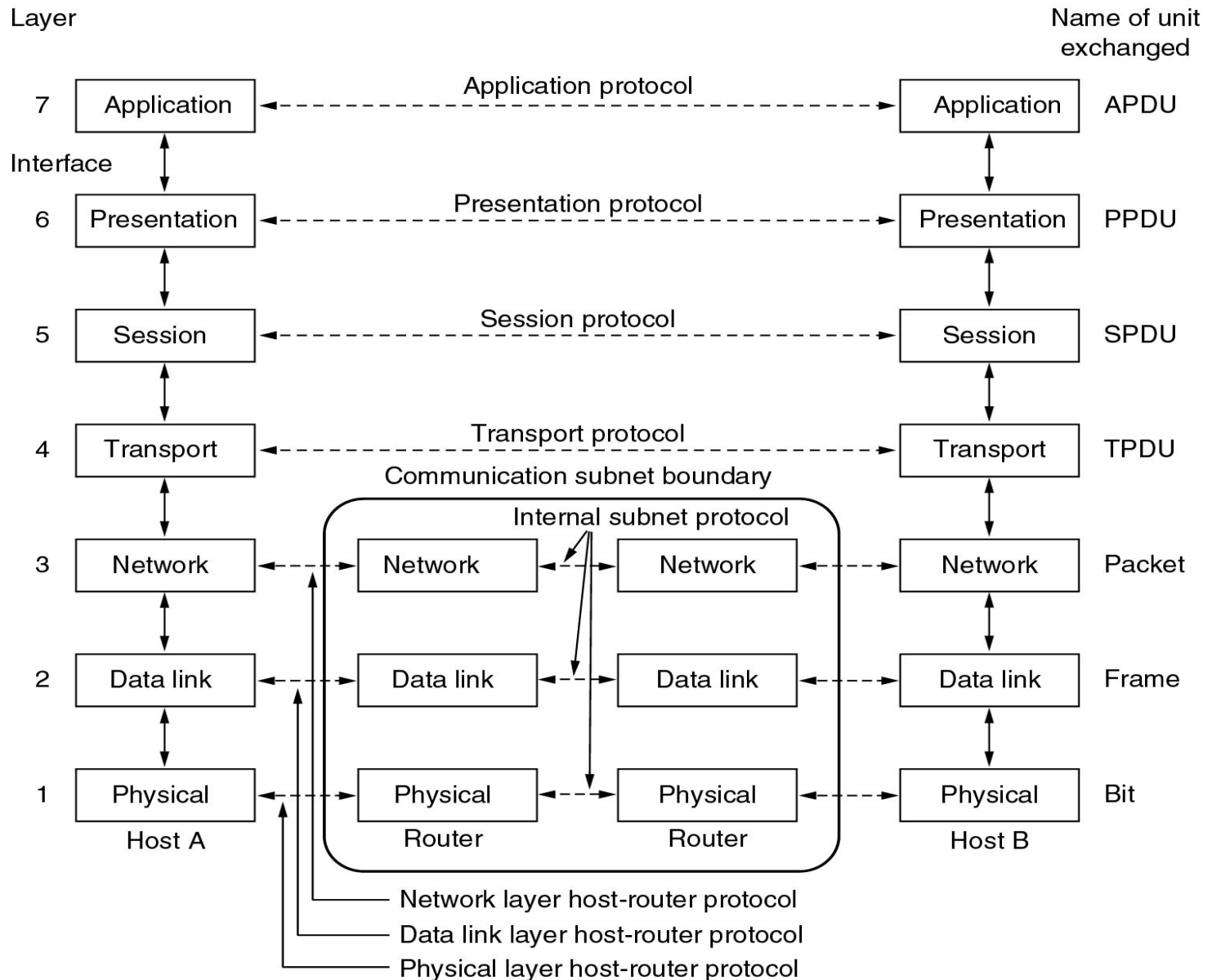


Reference Models

- The OSI Reference Model
- The TCP/IP Reference Model
- A Comparison of OSI and TCP/IP
- A Critique of the OSI Model and Protocols
- A Critique of the TCP/IP Reference Model

Reference Models

The OSI
reference
model.



Reference Models (2)

OSI

TCP/IP

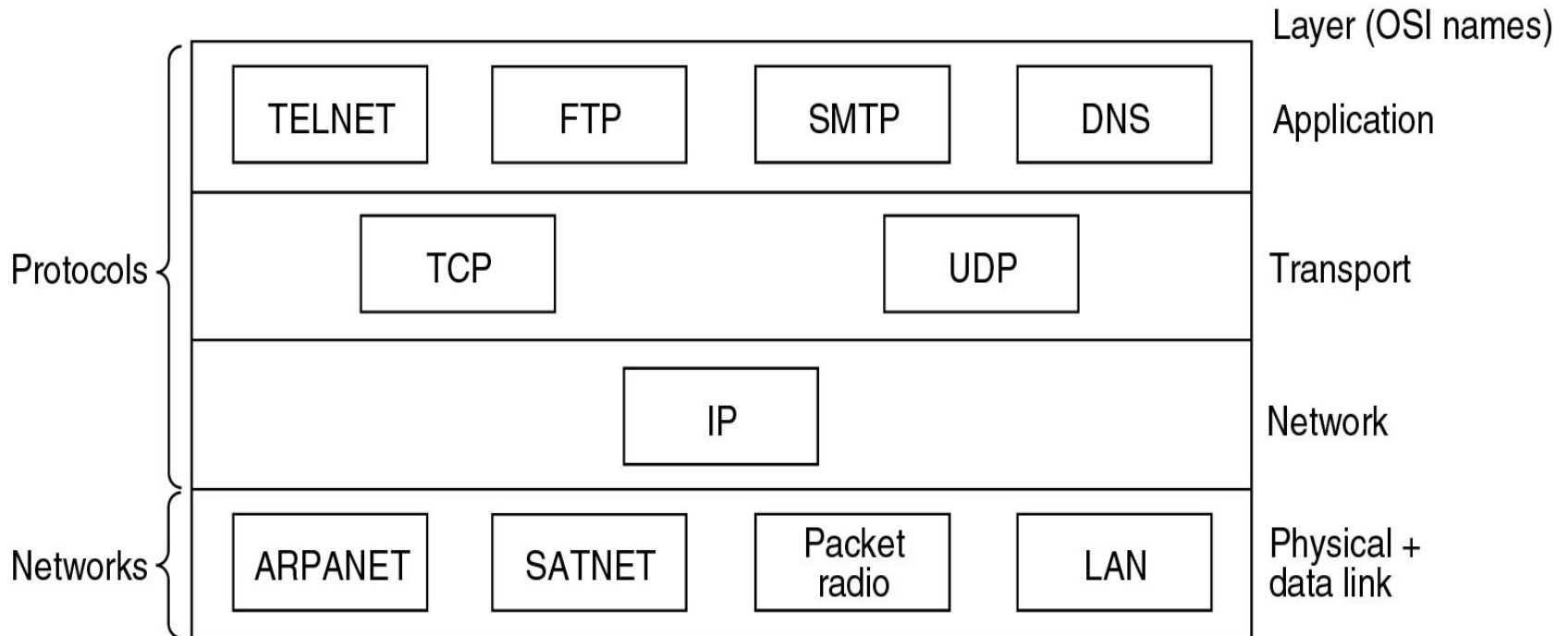
- The⁷

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data link
1	Physical

Application
Transport
Internet
Host-to-network

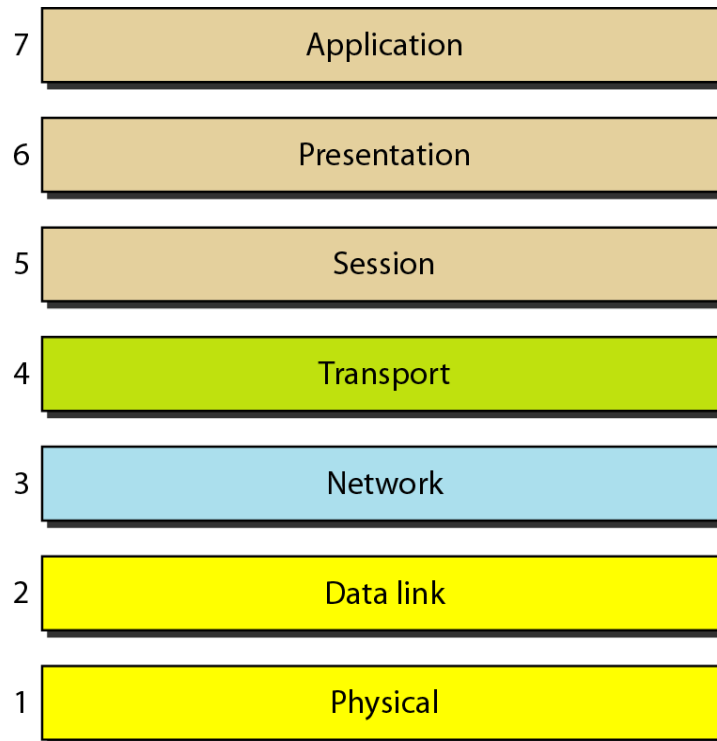
Not present
in the model

Reference Models (3)



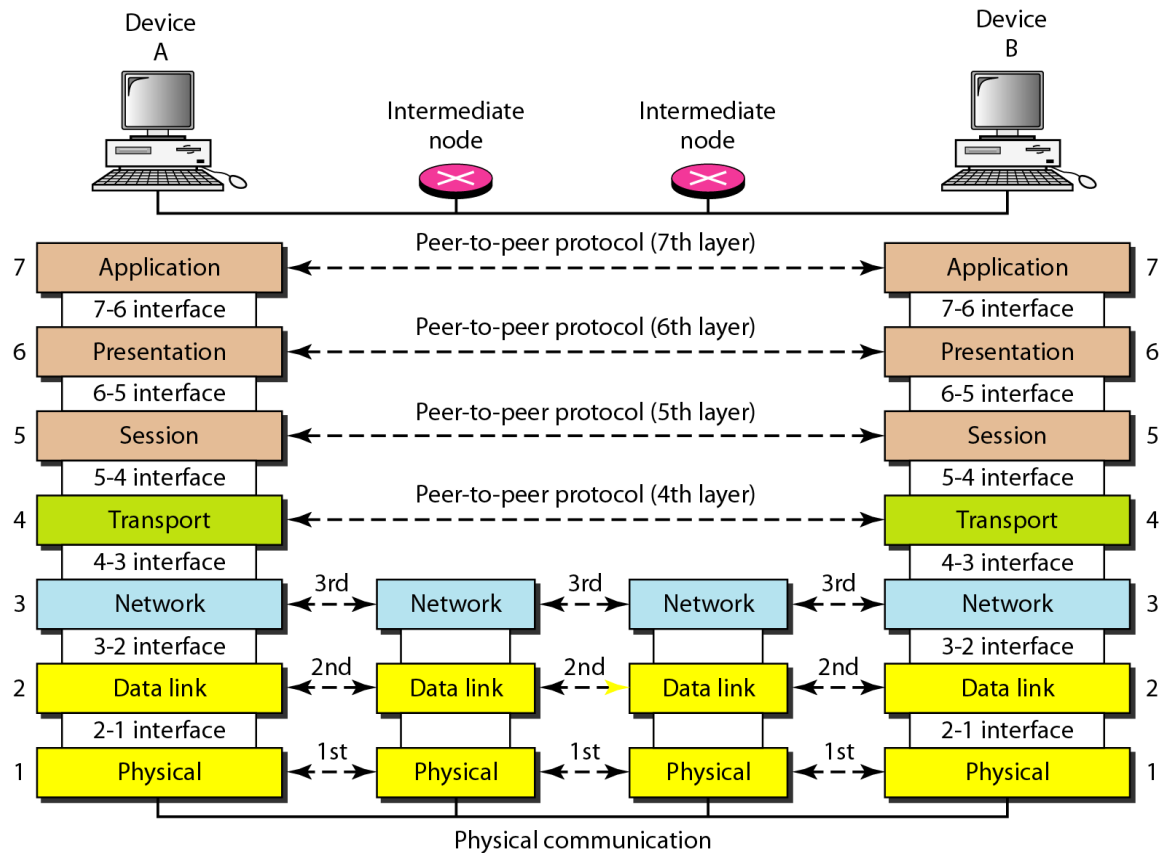
OSI Model

ISO is the organization. OSI is the model



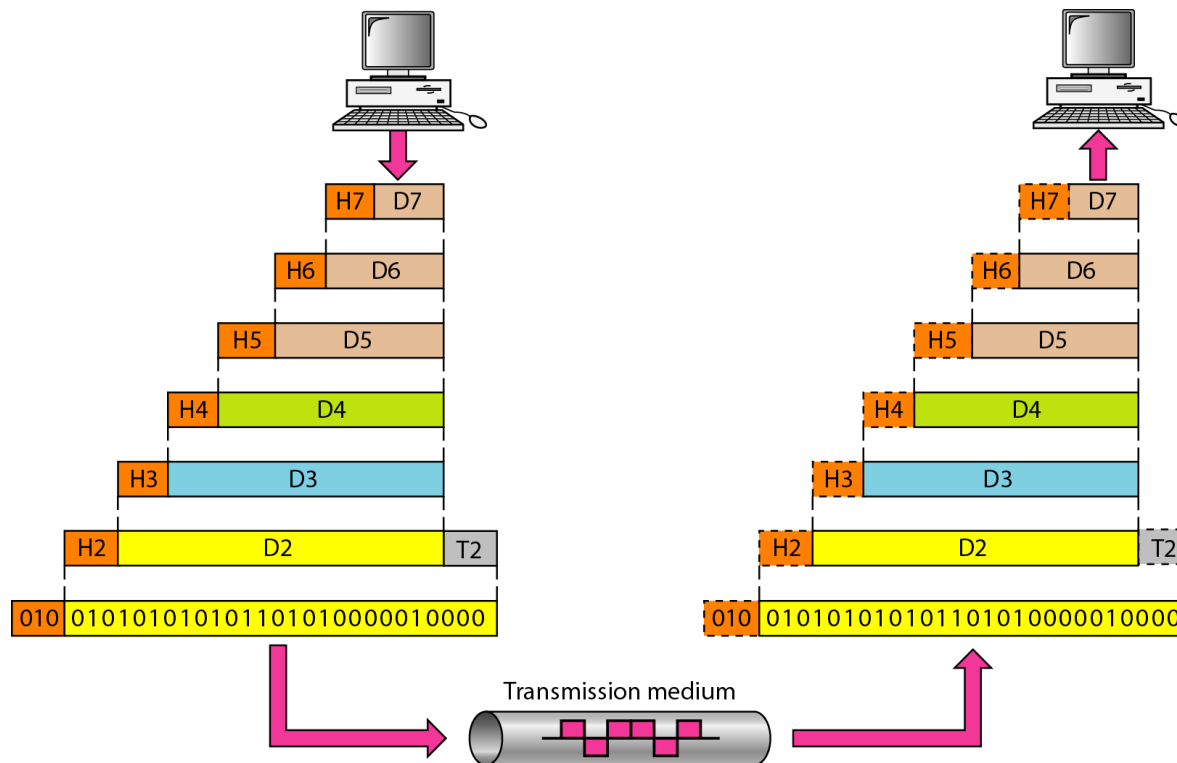
Interaction between layers in the OSI model

Layer and interface



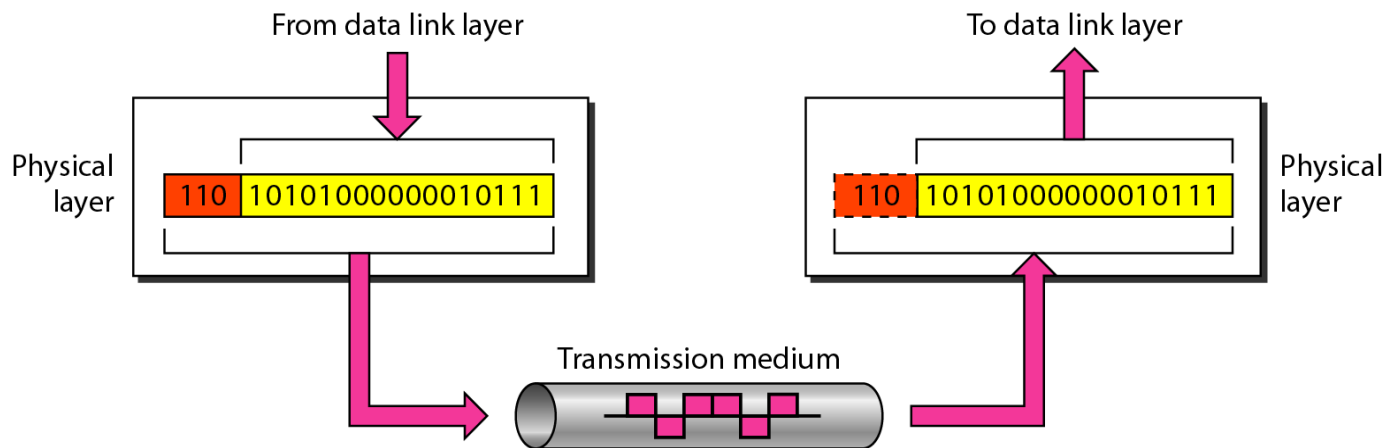
An exchange using the OSI model

Encapsulation with header and possibly trailer



Physical Layer

- The physical layer is responsible for movements of individual bits from one hop (node) to the next
- Mechanical and electrical specification, the procedures and functions

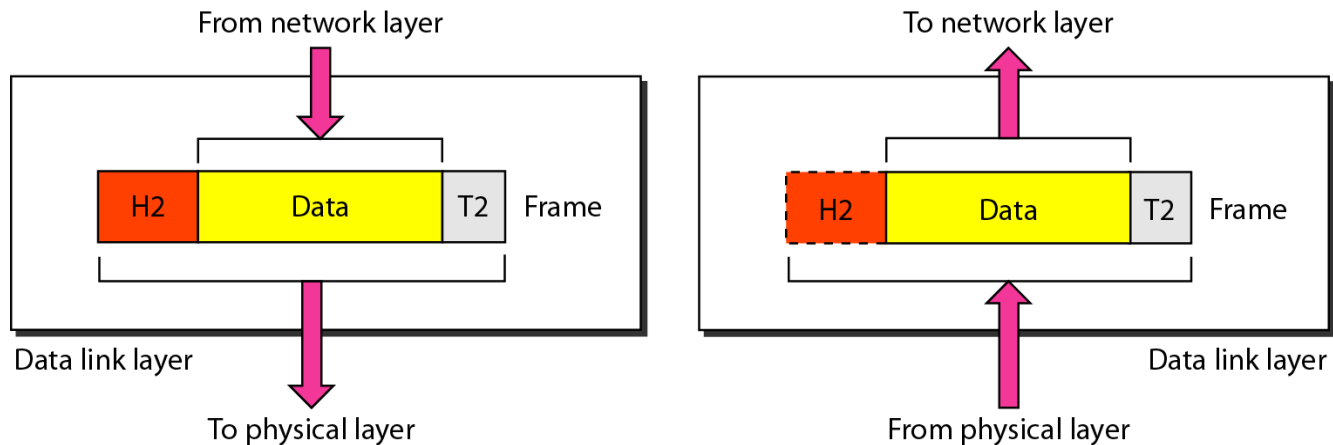


Physical Layer: Duties

- Physical characteristics of interfaces and media
- Representation of bits
- Data rate
- Synchronization of bits
- Line configuration
- Physical topology
- Transmission mode

Data Link Layer

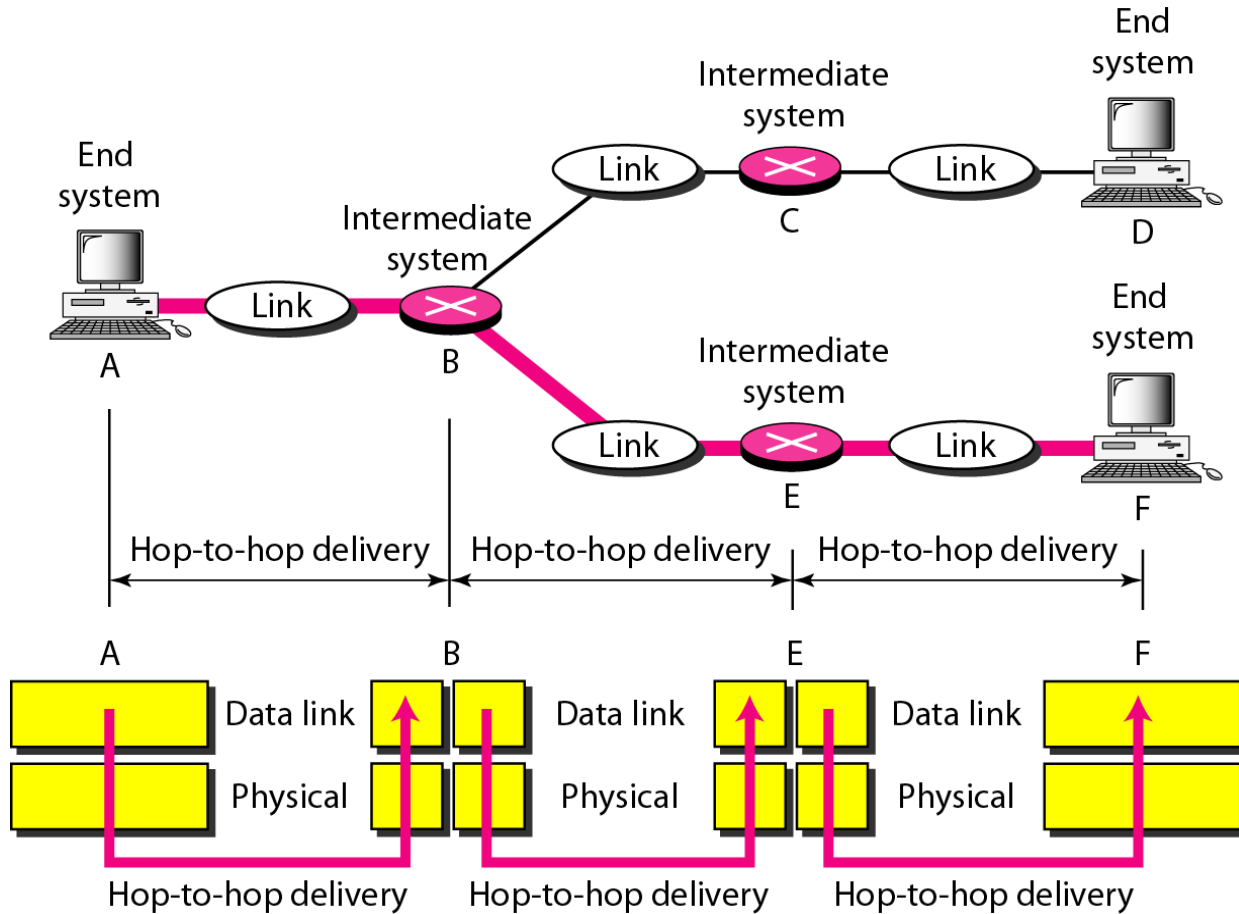
- The data link layer is responsible for moving frames from one hop (node) to the next
- Transform the physical layer to a reliable (error-free) link



Data Link Layer: Duties

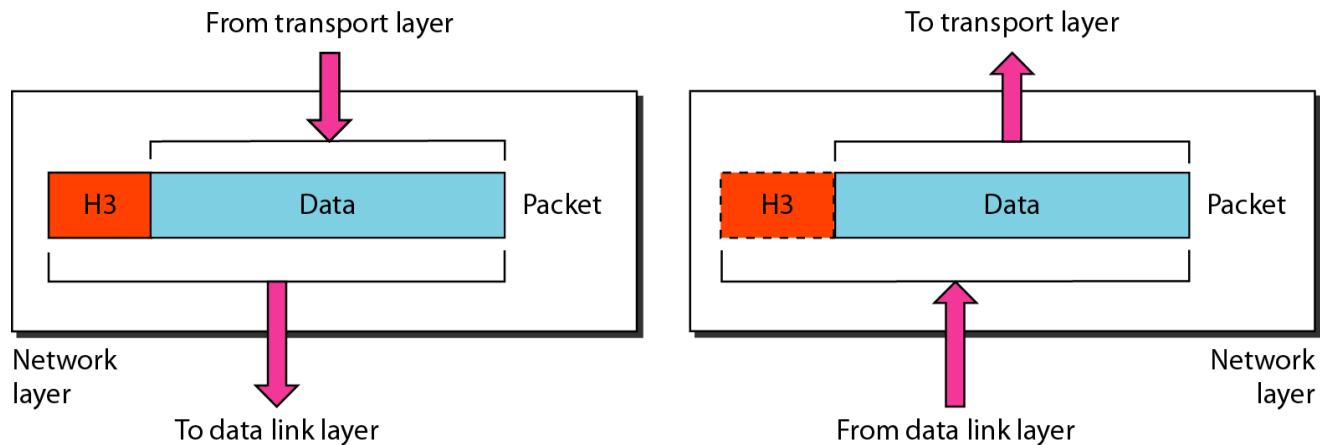
- Framing
- Physical addressing
- Flow control
- Error control

Hop-to-Hop Delivery



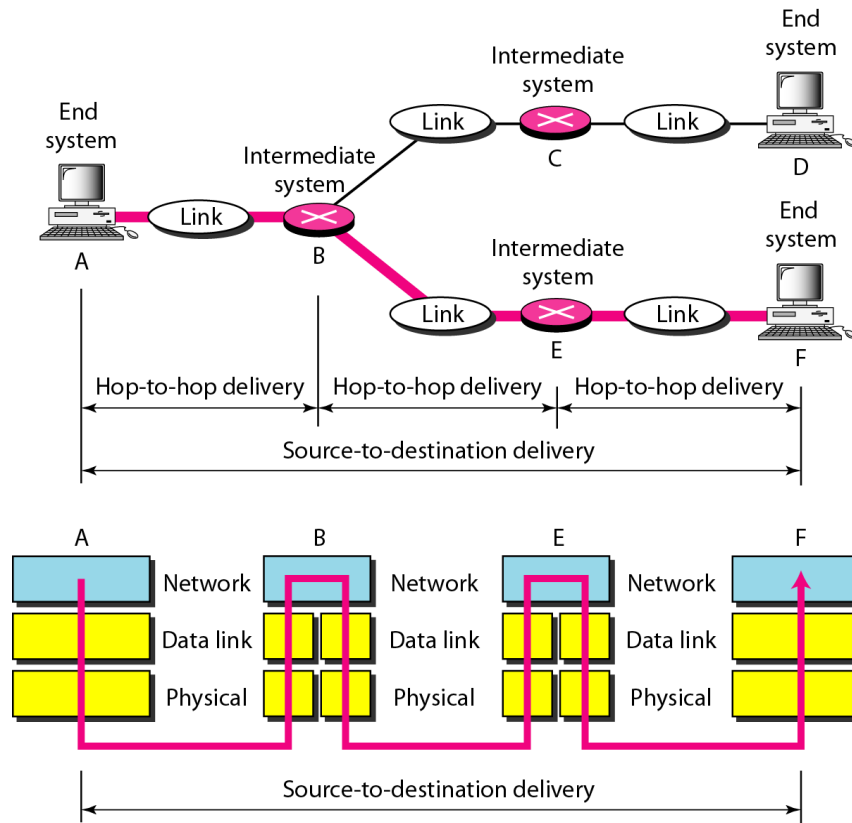
Network Layer

- The network layer is responsible for the delivery of packets from the source host to the destination host



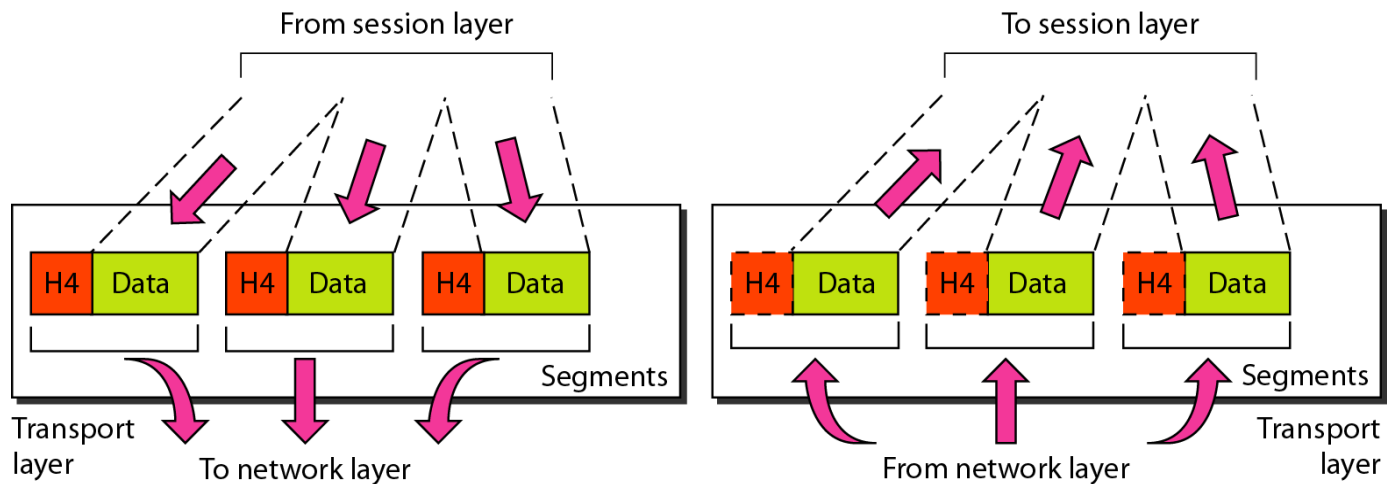
Network Layer: Duties

- Logical addressing and routing



Transport Layer

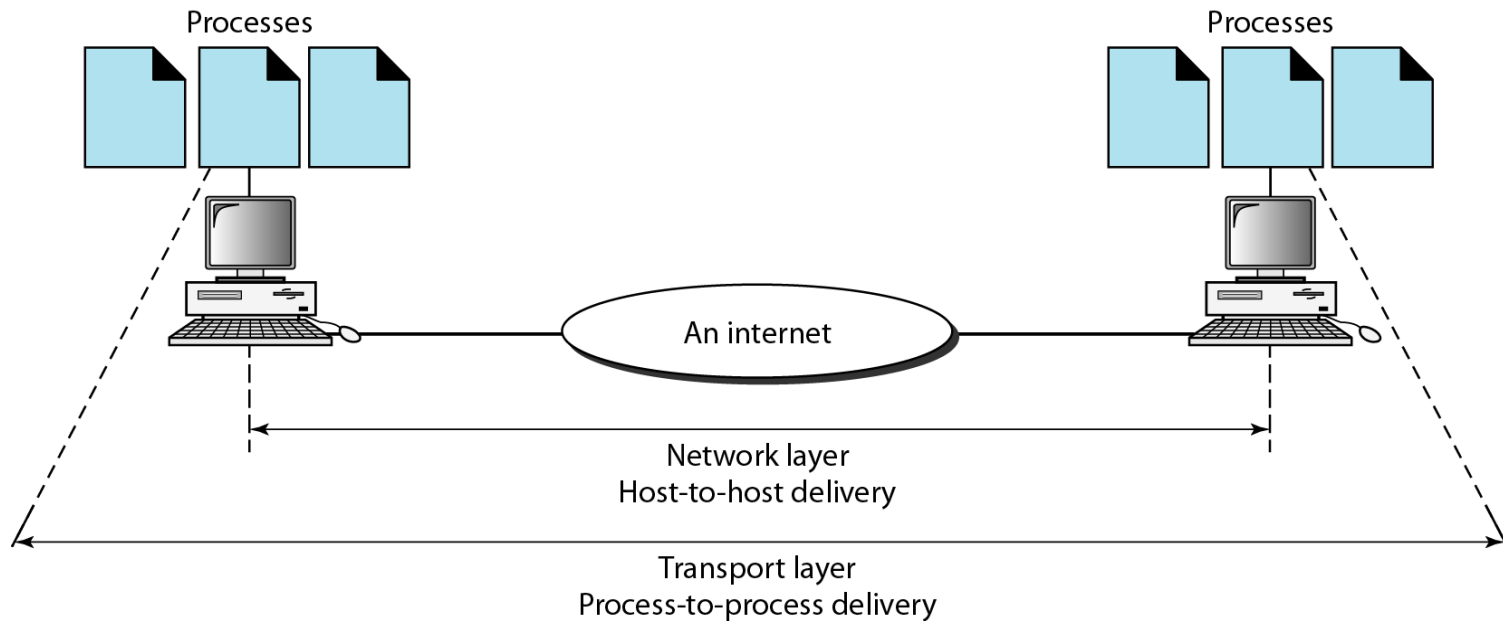
- The transport layer is responsible for delivery of a message from one process to another



Transport Layer: Duties

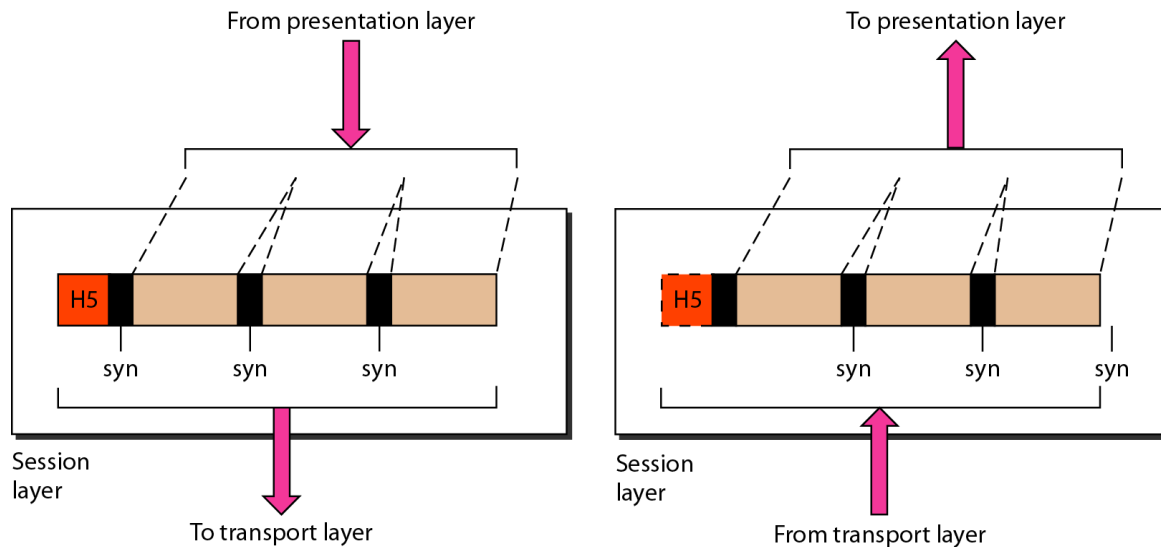
- Service-point (port) addressing
- Segmentation and reassembly
- Connection control
- Flow control
- Error control

Reliable Process-to-Process Delivery of a Message



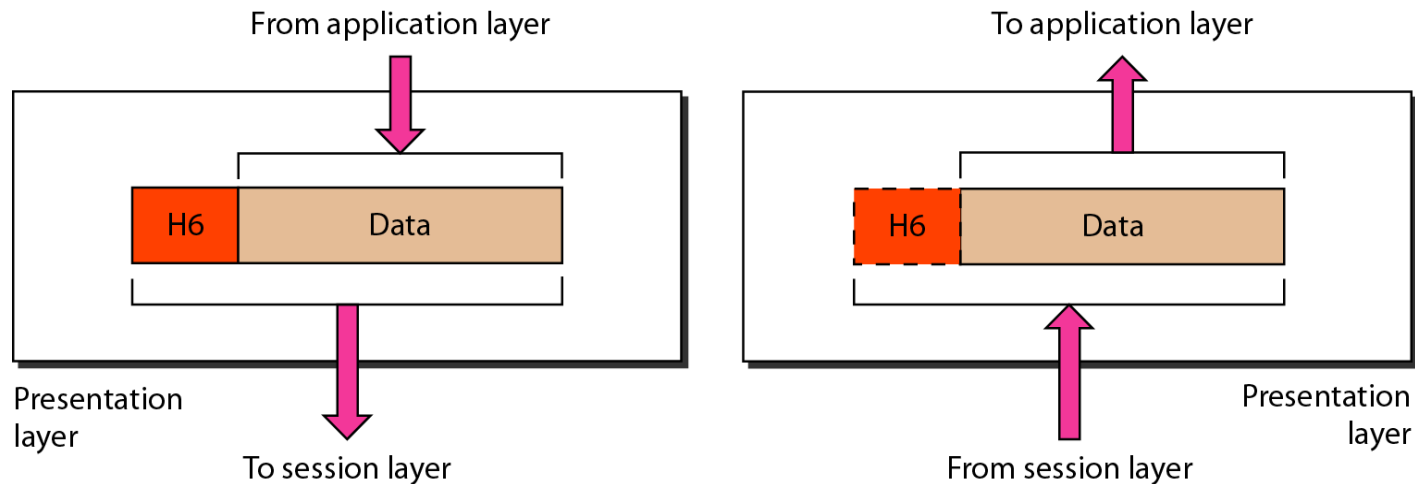
Session Layer

- Session layer is responsible for dialog control and synchronization



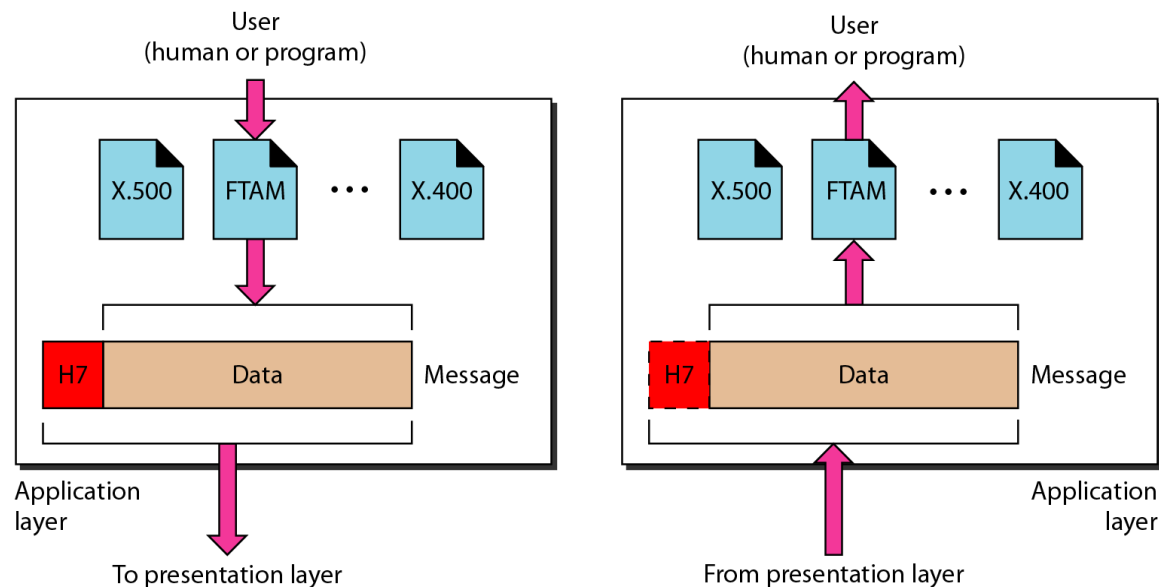
Presentation Layer

- Presentation layer is responsible for translation, compression, and encryption



Application Layer

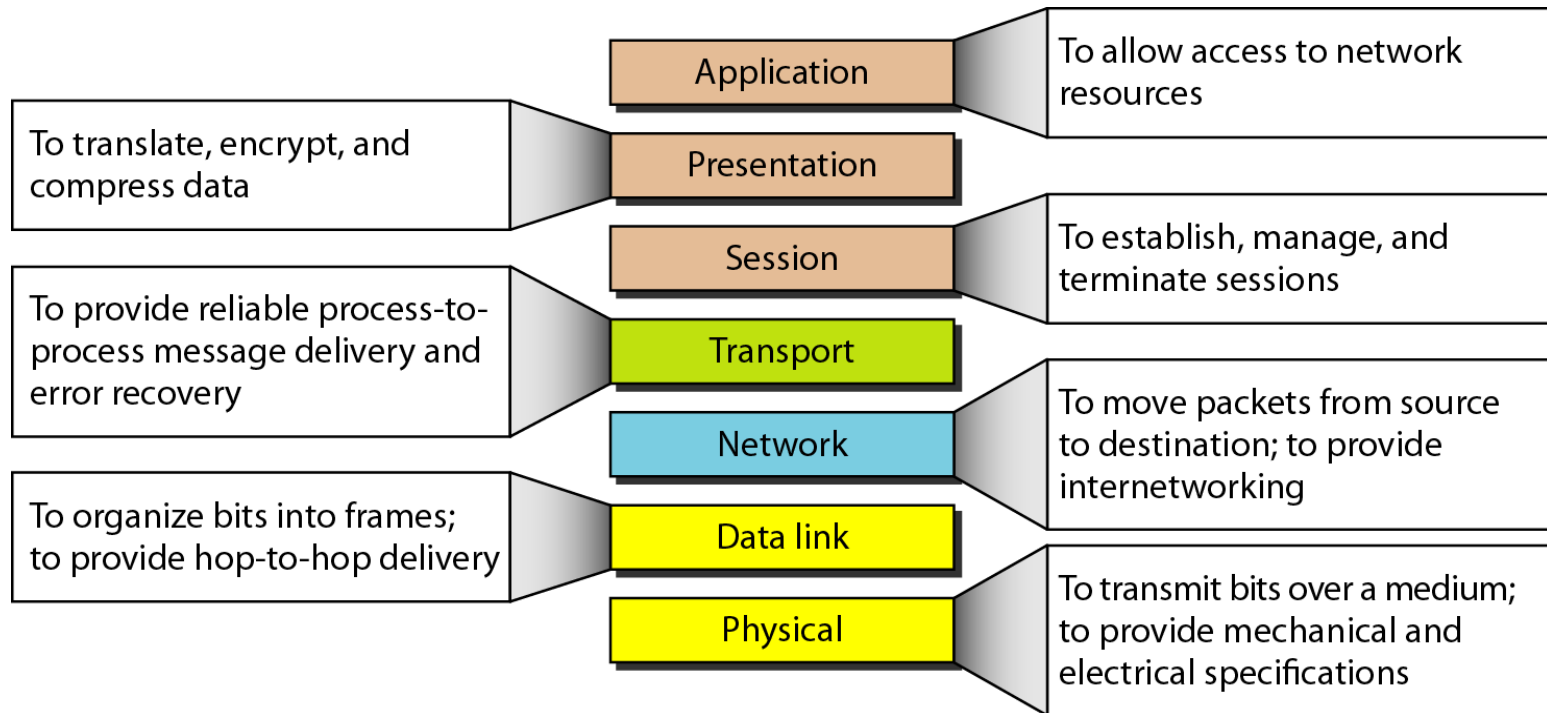
- Application layer is responsible for providing services to the user



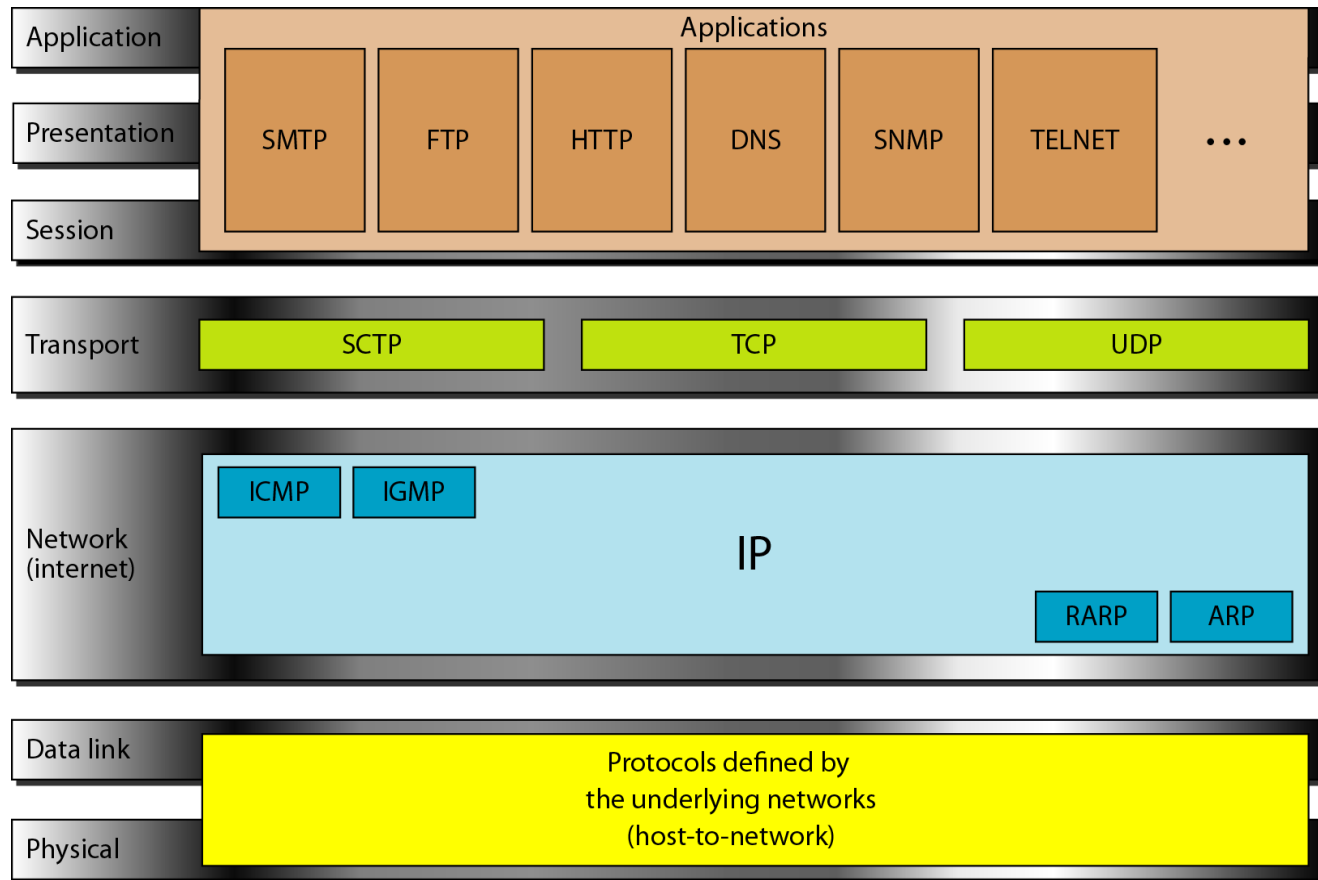
Application Layer: Services

- Network virtual terminal: It allows a user to log on to a remote host.
- Mail services
- File transfer, access, and management
- Directory services: provides distributed database sources and access for global information about various objects and services.

Summary of Layers



TCP/IP and OSI Model

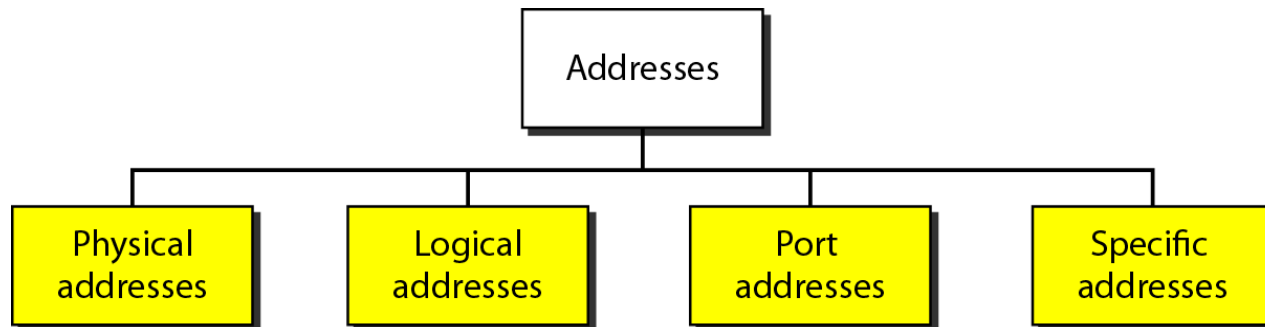


TCP/IP Protocol Suite

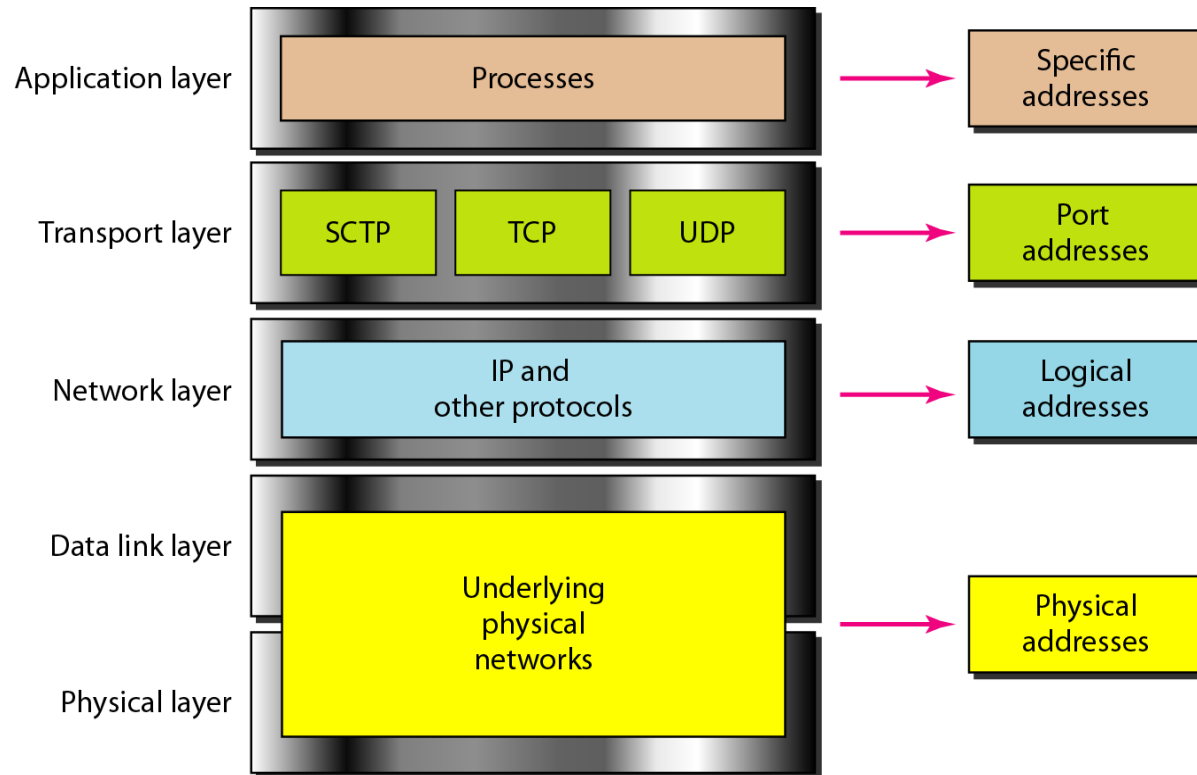
- Host-to-network : Physical and data link layer
 - No specific protocol
- Network layer
 - IP(Internet Protocol), ARP(Address Resolution Protocol), RARP(Reverse ARP), ICMP(Internet Control Message Protocol), IGMP(Internet Group Message Protocol)
- Transport layer
 - TCP(Transmission Control Protocol), UDP(User Datagram Protocol), SCTP(Stream Control Transmission Protocol),
- Application Layer
 - Combined session, presentation, and application layers

Addressing

- Four levels of addresses in TCP/IP protocols
- Physical (link), logical (IP, network), port, and specific addresses

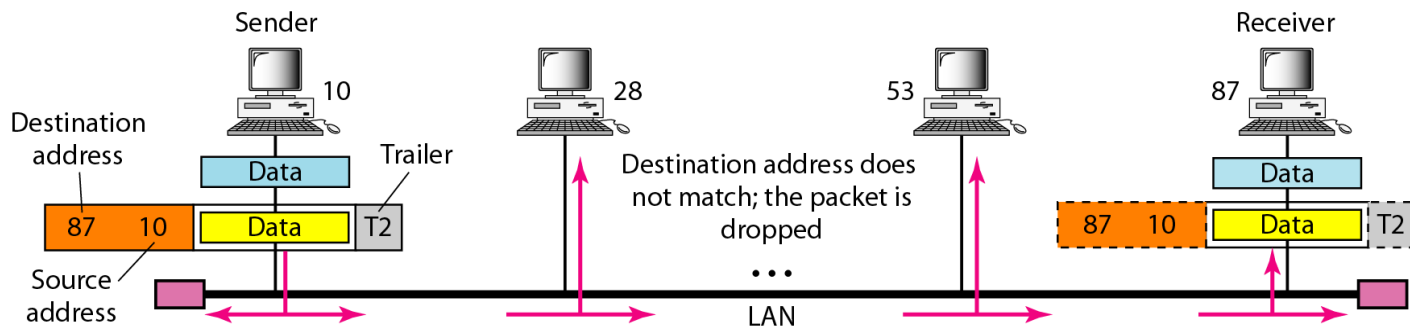


Relationship of Layers and Addresses



Physical Address

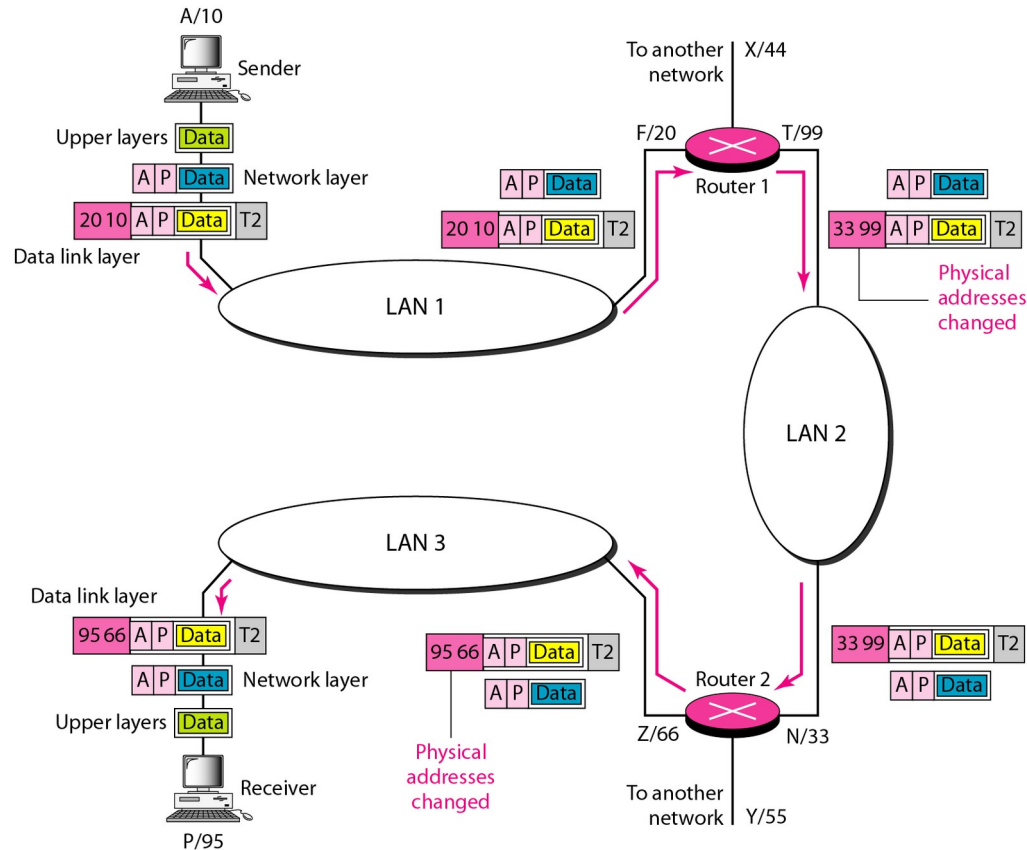
- A node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link (bus topology LAN). As the figure shows, the computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.



07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

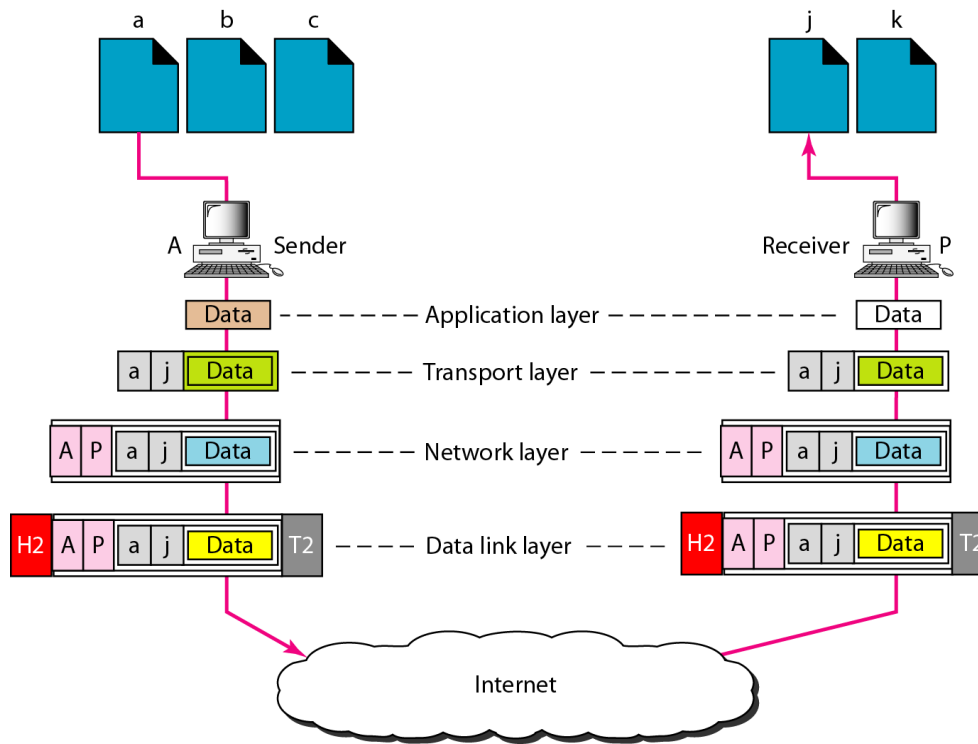
Logical (IP) Address



- The physical addresses will change from hop to hop, but the logical addresses usually remain the same

Port Address

- The physical addresses change from hop to hop, but the logical and port addresses usually remain the same



Specific Address

- Some application have user-friendly addresses that are designed for that specific address
- Example 1: e-mail address: kchung@kw.ac.kr
 - Defines the recipient of an e-mail
- Example 2: URL (Universal Resource Locator, now it is renamed as Uniform Resource Locator) :
www.kbs.co.kr
 - Used to find a document on the WWW

The physical layer

Physical Layer

The purpose of the physical layer is to transport a raw bit stream from one machine to another.

Various physical media can be used for the actual transmission.

Transmission Media:

Media are grouped into guided media and unguided media.

Guided media means waves are guided along solid medium.

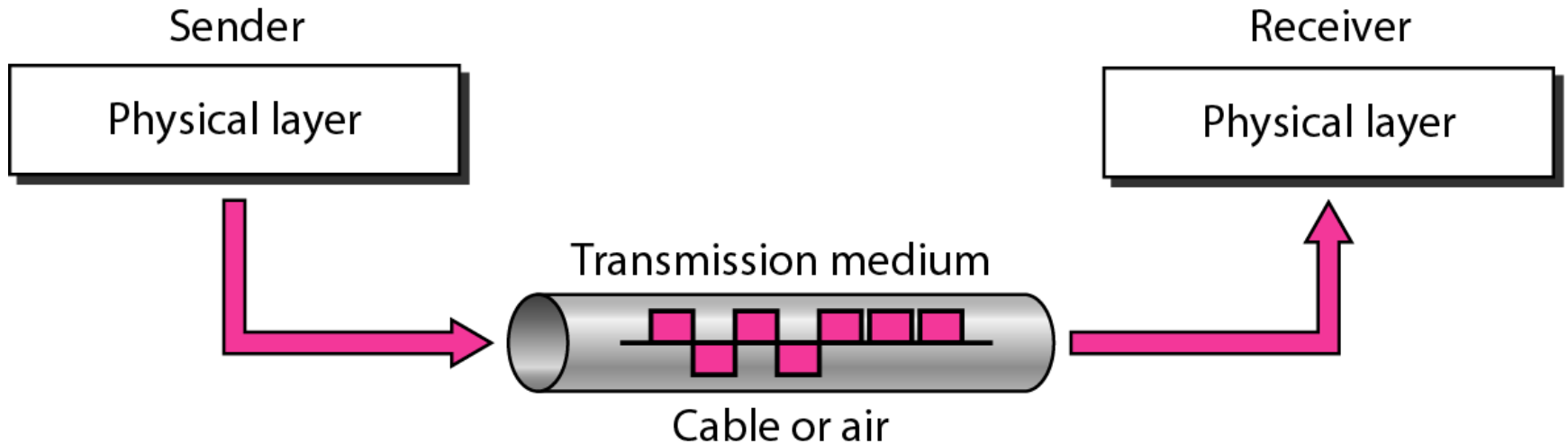
Ex: copper wire, fiber optics.

Unguided media is a media doesn't use any physical connectors between the two devices communicating.

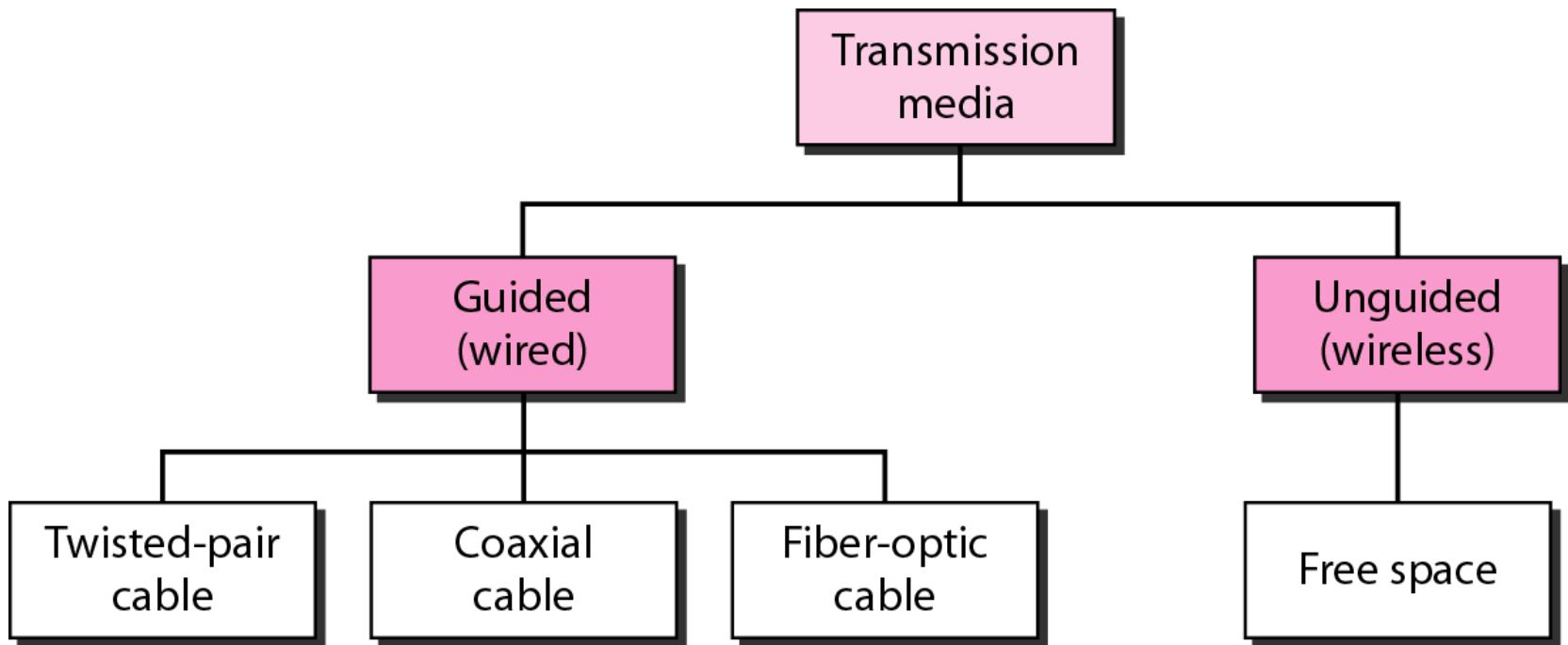
Usually the transmission is send through the atmosphere.

Ex: radio waves, micro waves, Infrared waves

The physical layer and transmission Medium



Classes of transmission media



Guided Transmission Data

- Magnetic Media
- Twisted Pair
- Coaxial Cable
- Fiber Optics

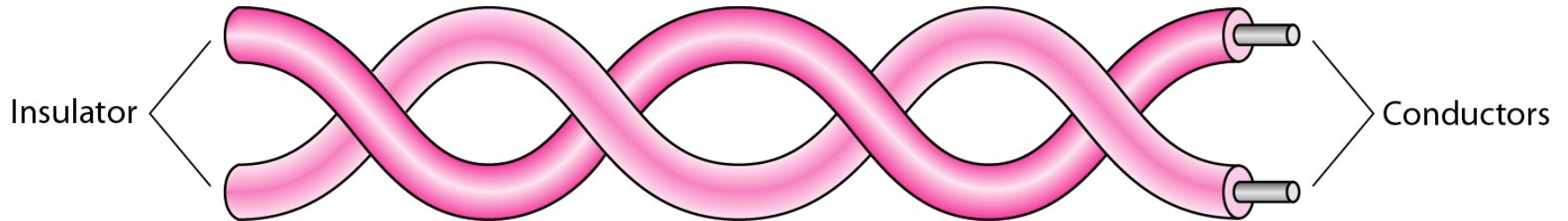
Magnetic Media

- One of the common way to transport data from one computer to another is to write them on to magnetic tape or removable media (DVD) physically transport the tape or DVD to the destination machine and read them.
- The tape densities are increasing.
- It is not suitable in applications, where on-line connection is needed

Twisted Pair

- Oldest transmission media
- A twisted pair consists of two insulated copper wires typically about 1mm thick. The wires are twisted together in a helical form.
- Twisting reduces interference (two parallel wires constitute a simple antenna; a twisted pair does not.)
- Cheap medium
- Commonly used for communications within buildings and in **telephone networks**
- For longer distance repeaters are needed.
- Twisted pairs can be used for transmitting either analog or digital signals.
- Produced in unshielded (UTP) and shielded (STP) forms, and in different performance categories.
- Cables may hold hundreds of pairs. Neighbor pairs typically have different twist lengths to reduce crosstalk.
- Category 5 has more twists per centimeter.

Twisted-pair cable



Twisted Pair



(a)

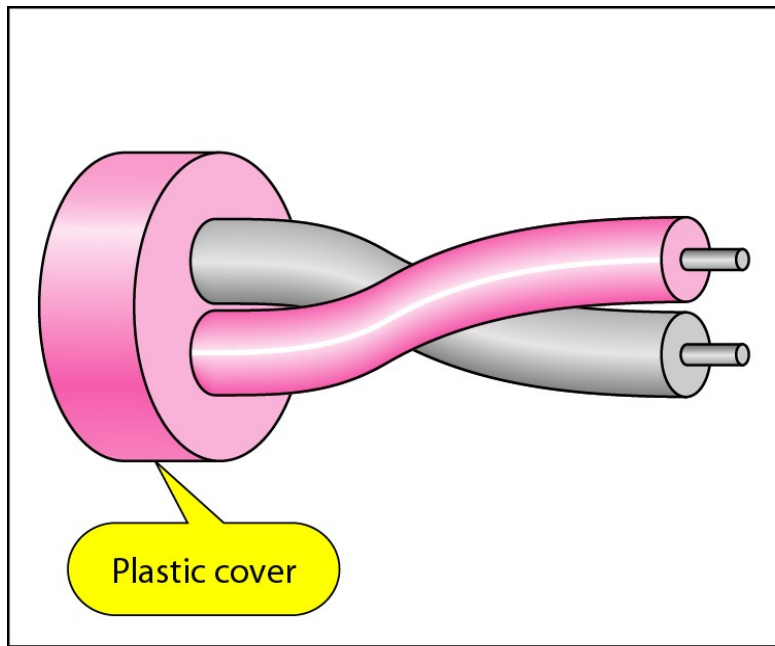


(b)

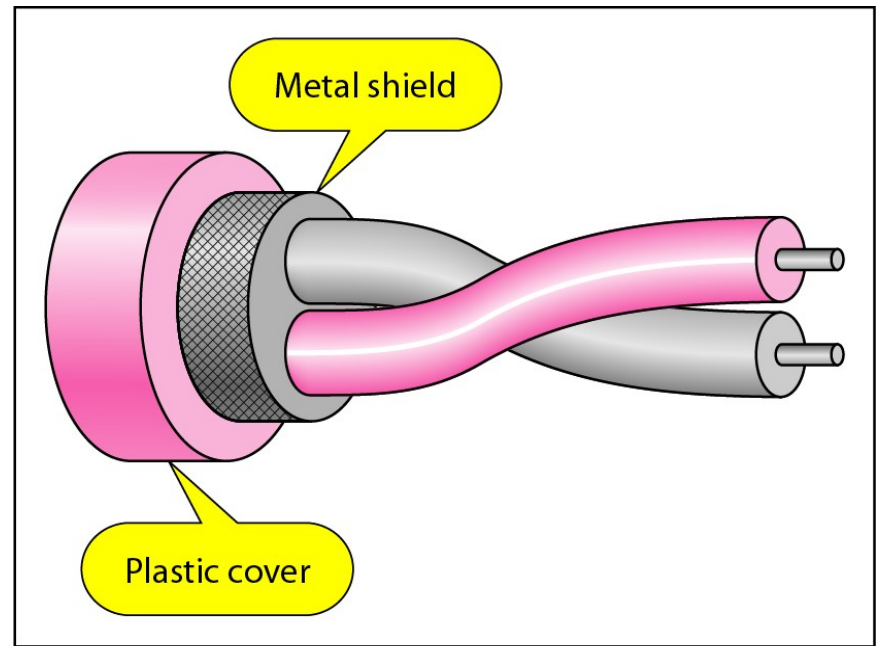
(a) Category 3 UTP.

(b) Category 5 UTP.

UTP and STP cables



a. UTP



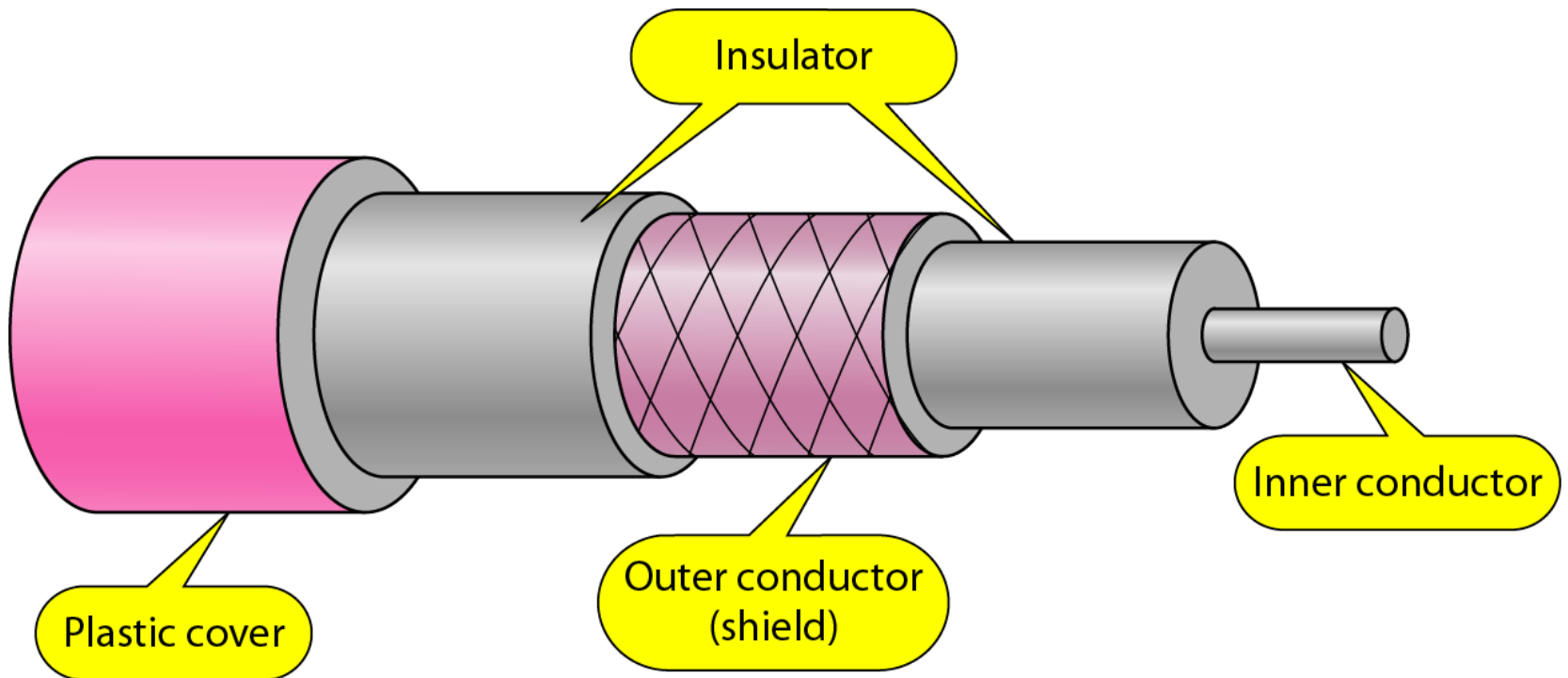
b. STP

Coaxial Cable

Pair of conductors separated by insulation

- It has better shielding than twisted pair
- Offers longer distances and better speeds than twisted pair, due to better shielding.
- Used for **cable TV and local-area networks**. Had been widely used in telephone systems, but optical fiber is now assuming this task.
- **Baseband Coaxial Cable** 50-ohm cable, commonly used for digital transmission. **Broadband Coaxial Cable** 75-ohm cable, commonly used for analog transmission.
- A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating material. The insulator is encased by a cylindrical conductor, often as a closely woven braided mesh. The outer conductor is covered in a protective plastic sheath.
- Good combination of high bandwidth and excellent noise immunity.
- It is widely used for **cable television and metropolitan networks**.

Coaxial cable



Categories of coaxial cables

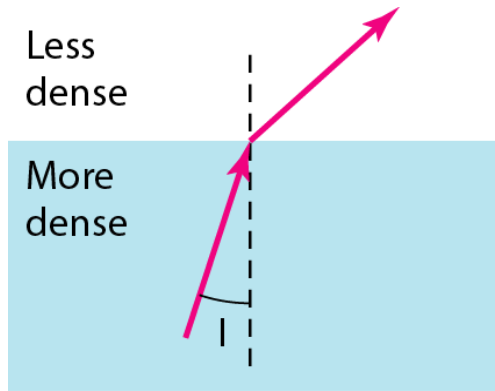
<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Fiber Optics

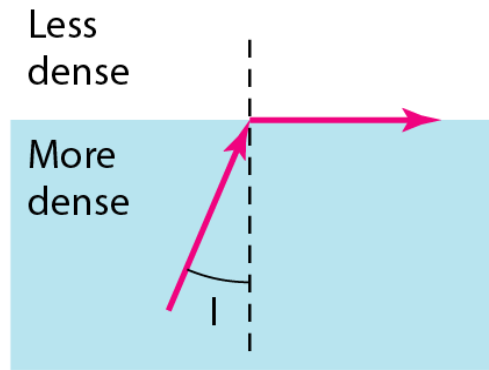
Optical Fiber

- Three key components: light source, transmission system, and a detector
- The detector generates an electric pulse when hit by light
- 1-a pulse of light; 0-missing pulse of light.
- optical rays travel in ultra-thin fiber of glass
- By attaching a light source to one end of an optical fiber and a detector to the other, we have a unidirectional data transmission system
- accepts an electrical signal converts to light at sending side transmits light pulses and then reconverts the light pulses to an electrical signal at the receiving end.
- A light ray incident at or above the critical angle is trapped inside the fiber.

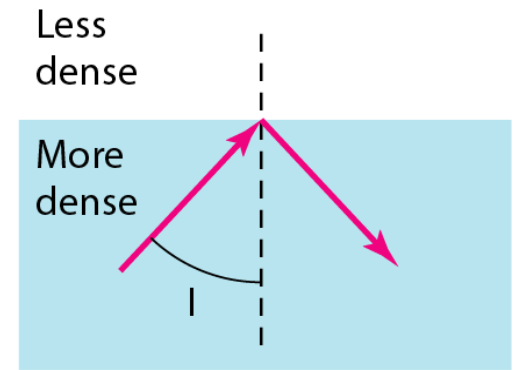
Bending of light ray



$i < \text{critical angle,}$
refraction

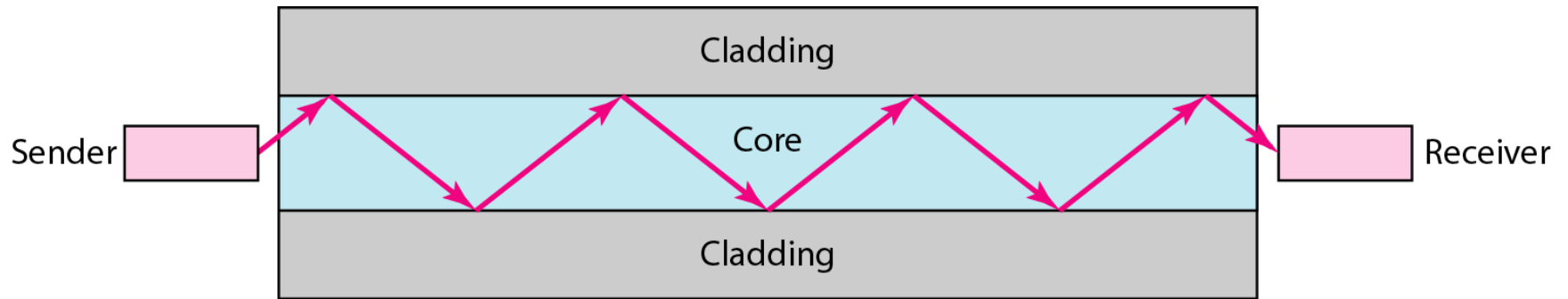


$i = \text{critical angle,}$
refraction



$i > \text{critical angle,}$
reflection

Optical fiber



Fiber optic cable

- Fiber optic cables are similar to coax, except without the braid.
- At the center is the glass core through which the light propagates.
- Multimode fiber, the core is typically 50 microns
- Single mode fiber, the core is 8 to 10 microns.
- The core is surrounded by a glass cladding, to keep all the light in the core.
- Next comes a thin plastic jacket to protect the cladding.
- Fibers are typically grouped in bundles, protected by an outer sheath.
- Fibers can be connected in three different ways.
 1. plugged in to fiber Sockets.
 2. Mechanical splices: clamp the two ends of fiber
 3. fusion splice: two pieces of fiber can be fused(melted) to form a solid connection

Fiber optic cable contd...

Two kinds of light sources are typically used to do the signaling. Electrical signals are converted into light.

- LED (Light Emitting Diode)
- semiconductor lasers

Light Detector

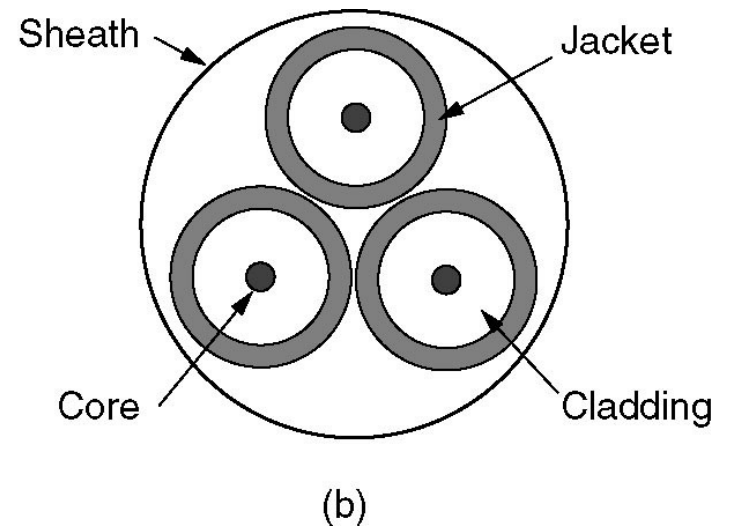
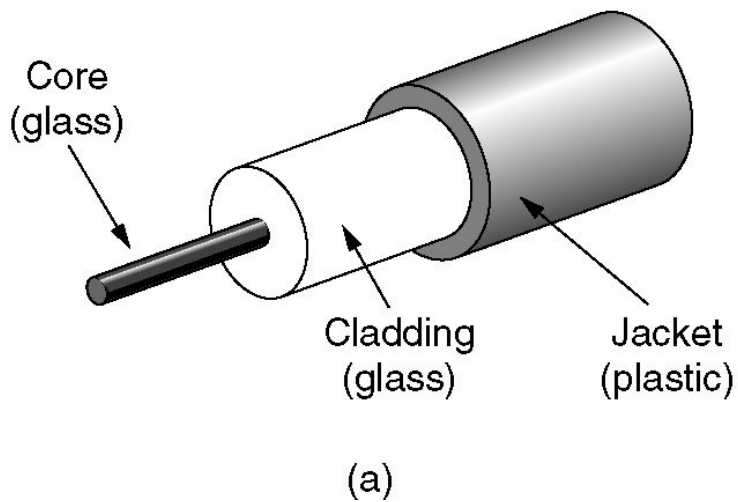
- Photo diode: it gives an electrical pulse when struck by light.

Fiber optics can be used for LANs as well as for long transmission.

Two types of interfaces are used.

- 1) passive interface
- 2) active interface

Fiber Cables



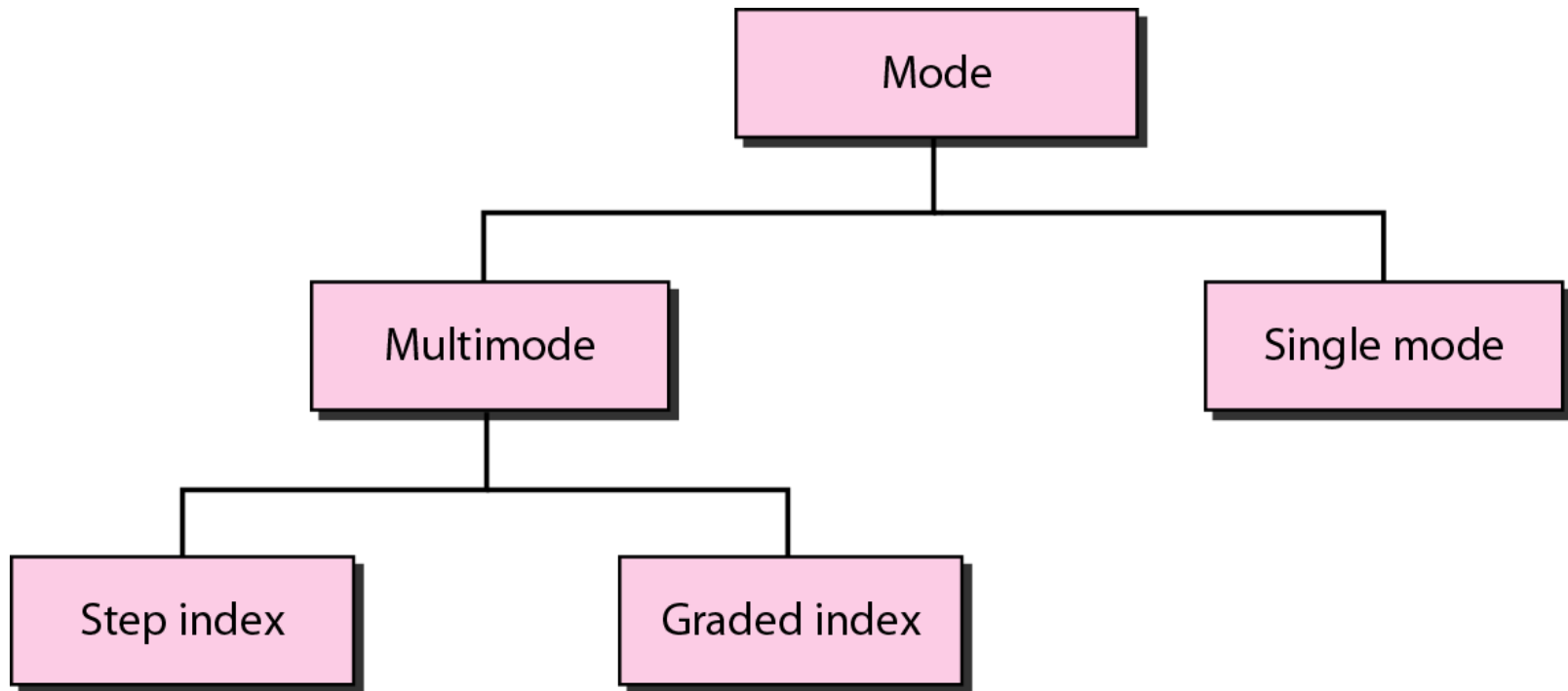
(a) Side view of a single fiber.

(b) End view of a sheath with three fibers.

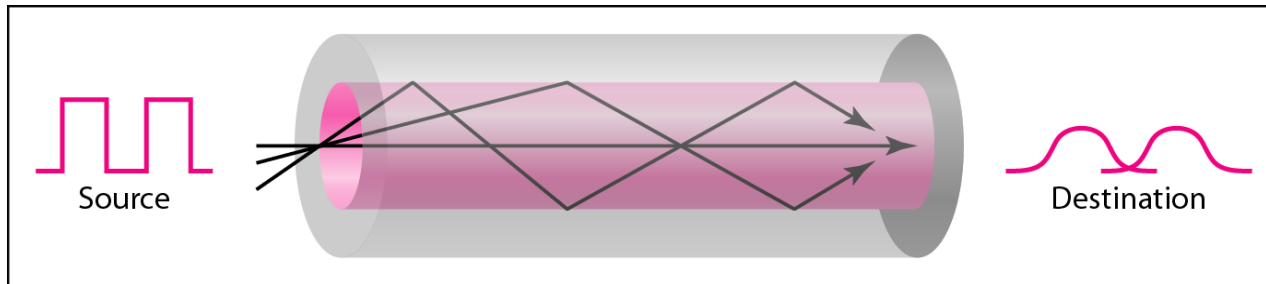
Fiber Optics contd..

- Many different rays will be bouncing internally around at different angles. Each ray is said to have a different mode, so a fiber having this property is called a **multimode fiber**.
- If the fiber's diameter is reduced to a few wavelengths of light, the light can propagate only in a straight line, without bouncing, this property is called a **single mode fiber**.
- Single mode fiber are more expensive but are widely used for longer distances. Transmit data at 50 Gbps for 100Km without amplification.

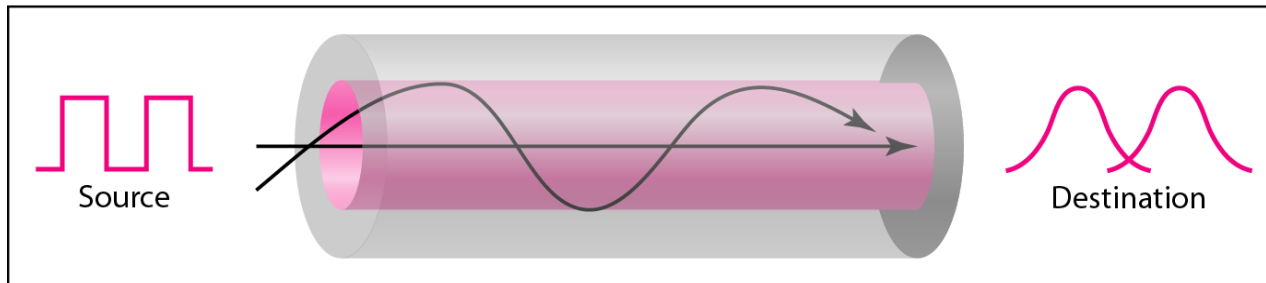
Propagation modes



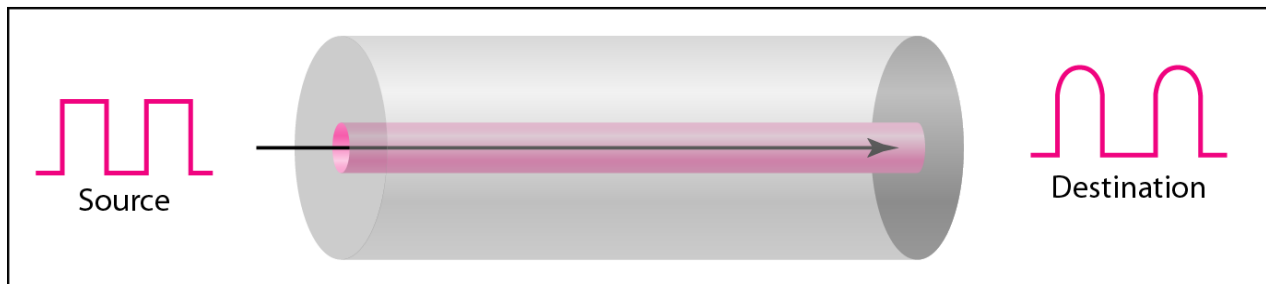
Modes



a. Multimode, step index

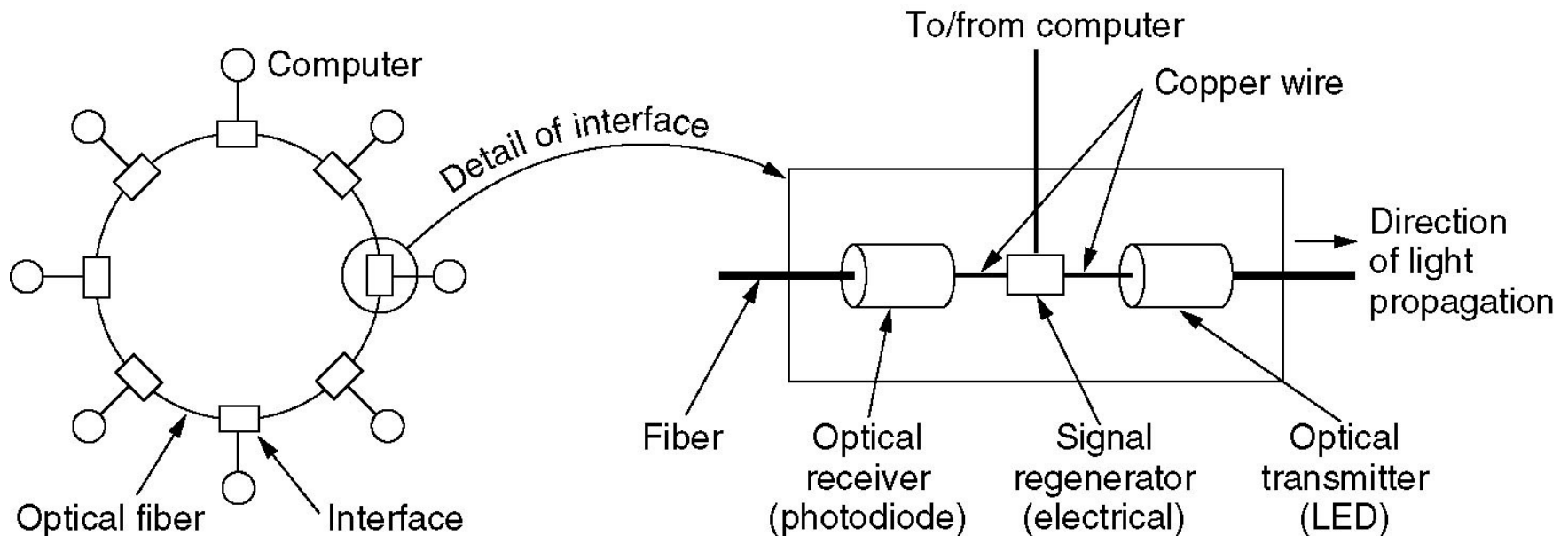


b. Multimode, graded index



c. Single mode

Fiber Optic Networks



A fiber optic ring with active repeaters.

Comparison between Fiber Optics and Copper Wire

Fiber optic

- 1) It can handle higher bandwidth than copper
- 2) Due to low attenuation repeaters are needed only about every 50 km long line, whereas copper wire every 5 km
- 3) Cost saving as less repeaters are needed
- 4) Not being affected by electromagnetic interference or power failures. It is not affected by corrosive chemicals
- 5) Fibers do not leak light and are quite difficult to tap, excellent security against potentially wire tappers
- 6) Thin and light weight
- 7) Lower installation cost
- 8) Few skilled engineers having fiber technology are available
- 9) Fibers can be damaged easily by being bent too much.
- 10) Two way communication requires two fibers
- 11) Fiber interfaces are costly
- 12) Long distance fiber is good choice

7-2 UNGUIDED MEDIA: WIRELESS

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.

Topics discussed in this section:

Radio Waves Microwaves

Infrared

Figure 7.17 *Electromagnetic spectrum for wireless communication*

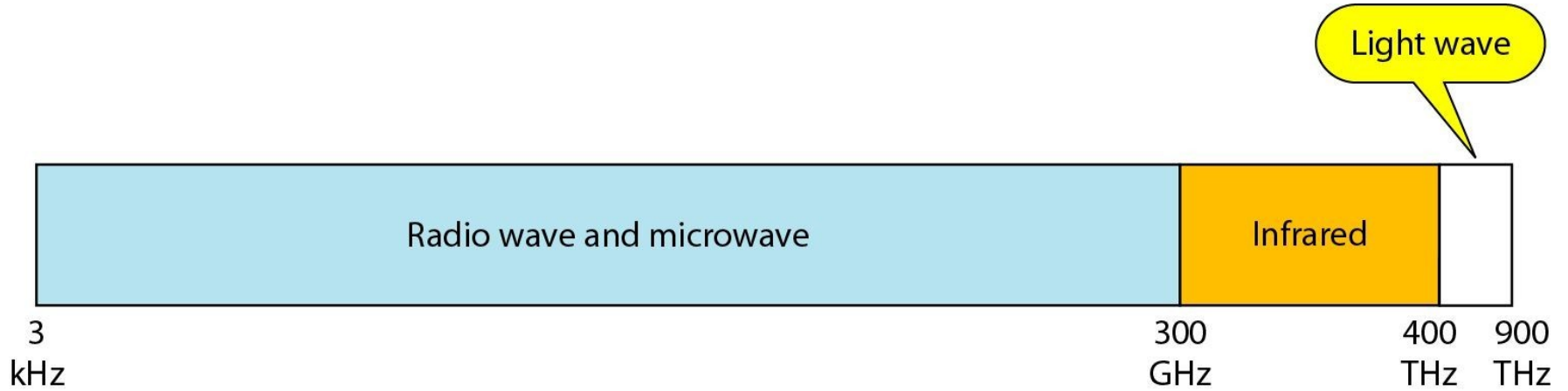


Figure 7.18 *Propagation methods*

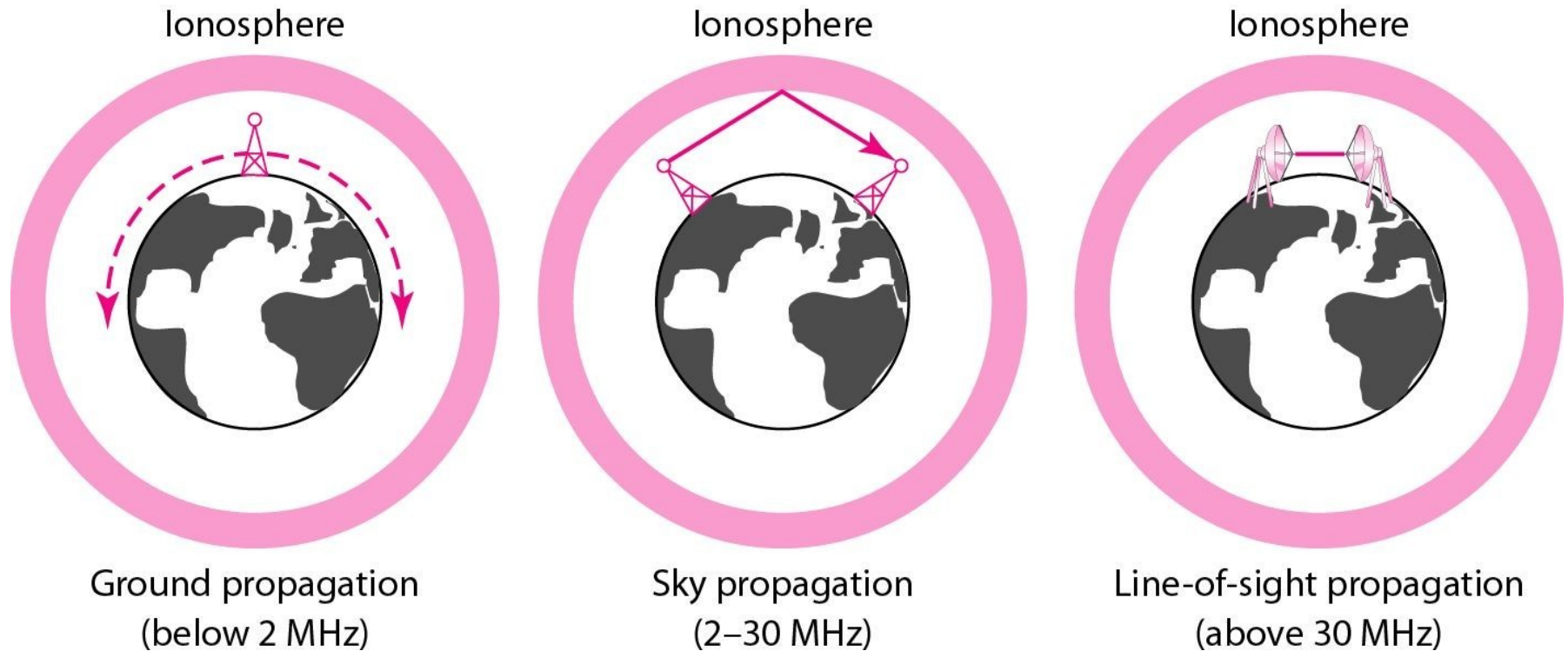
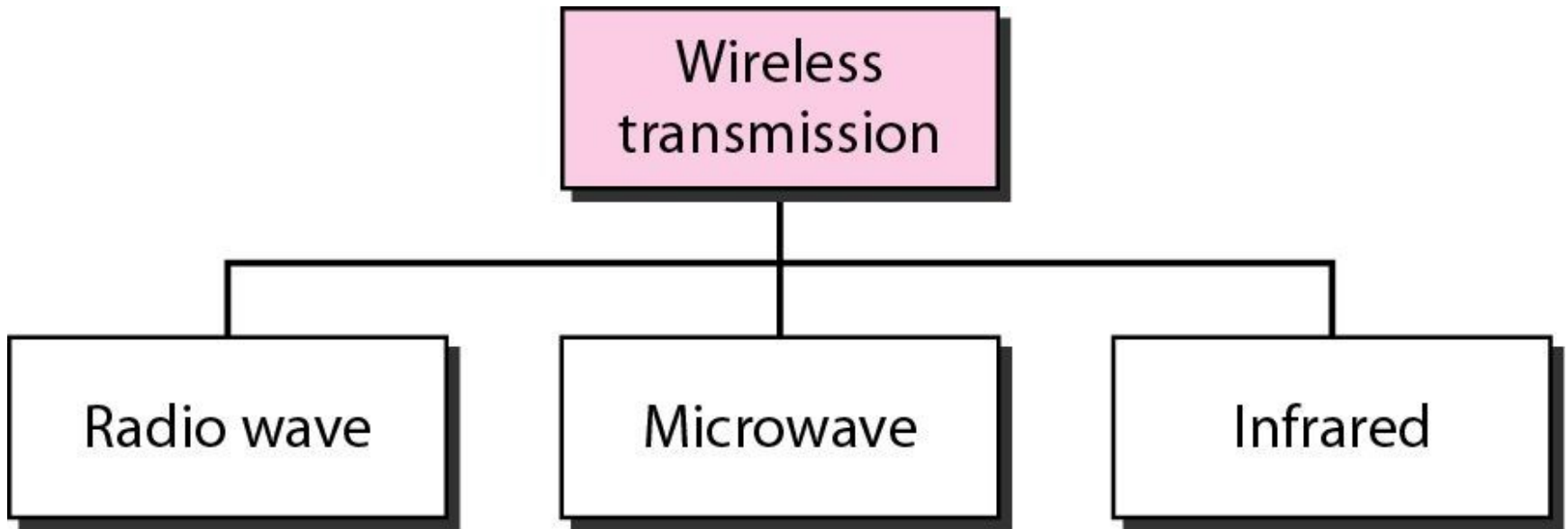


Table 7.4
Bands

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

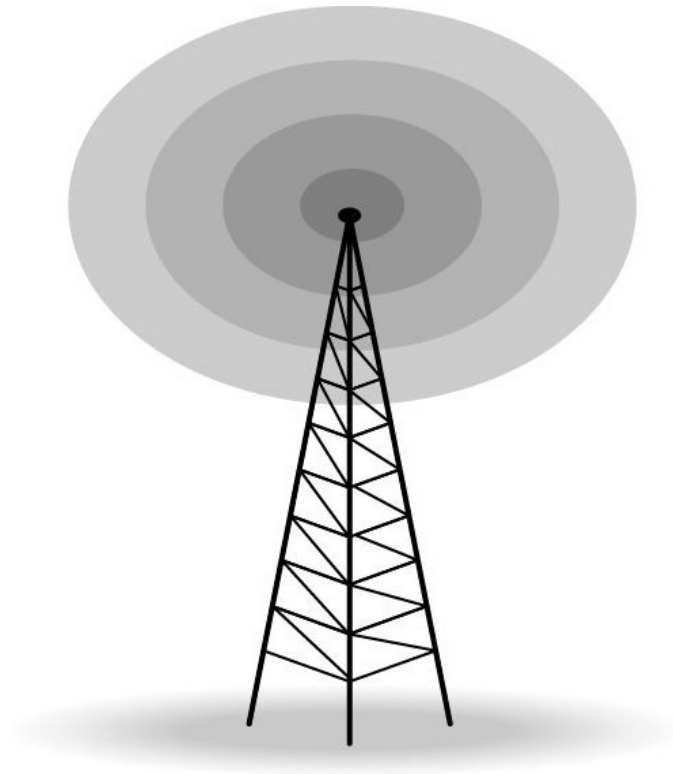
Figure 7.19 *Wireless transmission waves*



Note

Radio waves are used for multicast communications, such as radio and television, and paging systems. They can penetrate through walls. Highly regulated. Use omnidirectional antennas

Figure 7.20 *Omnidirectional antenna*



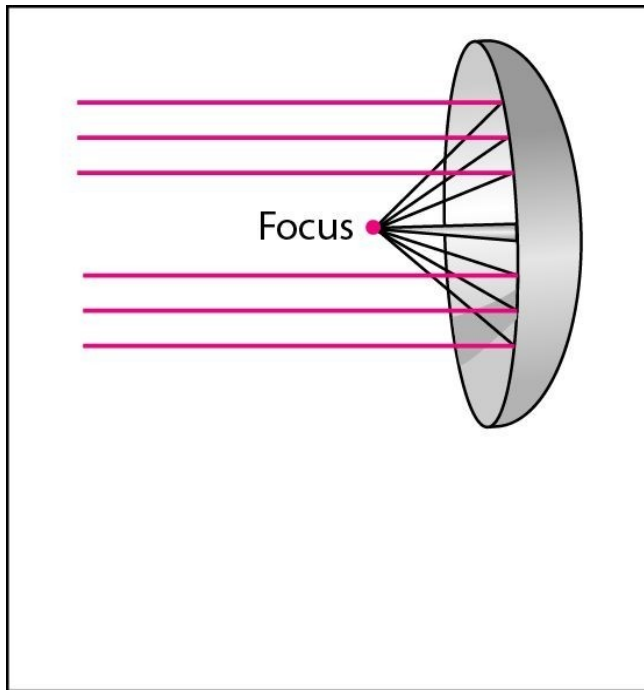
Note

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

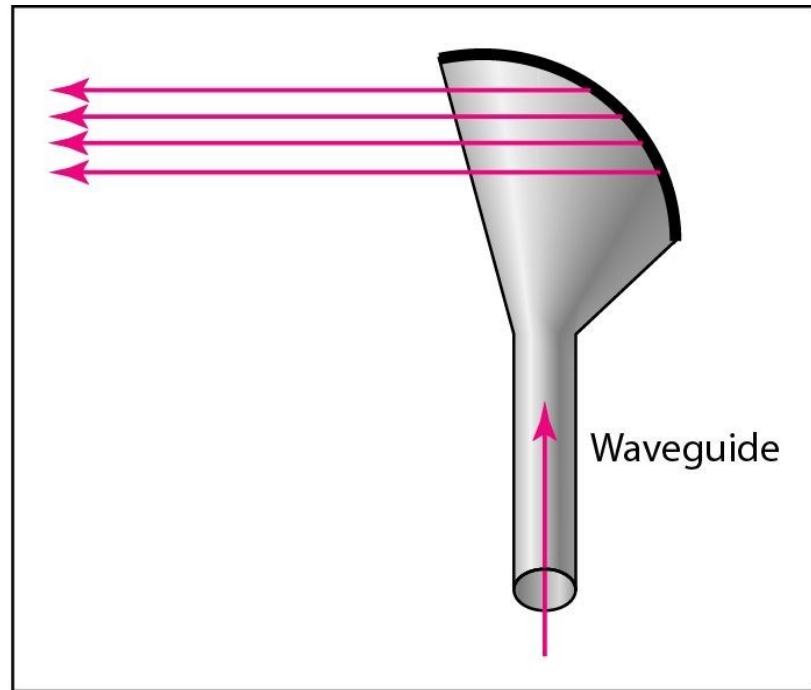
Higher frequency ranges cannot penetrate walls.

Use directional antennas - point to point line of sight communications.

Figure 7.21 *Unidirectional antennas*



a. Dish antenna



b. Horn antenna

Note

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Wireless Channels

- Are subject to a lot more errors than guided media channels.
- Interference is one cause for errors, can be circumvented with high SNR.
- The higher the SNR the less capacity is available for transmission due to the broadcast nature of the channel.
- Channel also subject to fading and no coverage holes.